

Efficient Video Super-Resolution for Fluorescence Microscopy

A Comparison of Transformer and GRU-Based
Models for Low-Latency SOFI Reconstruction

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Abstract

Capturing dynamic processes happening inside live cells is difficult due to the fact that many subcellular structures are completely hidden beyond diffraction blur. Super-resolution optical fluctuation imaging (SOFI) surpasses the optical diffraction limit using only a conventional widefield microscope, but it requires hundreds of frames and offline cumulant computation, making it too slow for real-time live-cell imaging. Recent work (RESURF) showed that a recurrent deep network, SOFI-MISRGRU, can approximate a second-order SOFI reconstruction from as few as eight frames at real-time speed. Its fusion mechanism is inherently sequential and aggregates frames through an unnormalised summation that does not generalise beyond the training frame count.

This thesis investigates whether a Transformer that fuses all frames in parallel through self-attention (TR-MISR) can improve upon this recurrent fusion in the structural fidelity, the number of frames the model can exploit, and the inference latency per output image. Quality is quantified with the resolution-scaled Pearson correlation (RSP) and a decorrelation-based resolution estimate, matching the RESURF evaluation, on a synthetic microtubule dataset spanning twelve blinking and signal-to-noise conditions.

Secondly, we modify SOFI-MISRGRU into a streaming architecture that emits one super-resolved image per incoming frame. Which improved per output latency as well as allowed the model to accumulate more frames improving reconstruction fidelity.

TR-MISR matches the 20-frame MISRGRU baseline while holding its quality under low SNR, slow-blinking acquisitions where MISRGRU degrades and hallucinates some filaments. Because cross-attention normalises over the input tokens, an eight-frame-trained TR-MISR can be evaluated on longer sequences without retraining, a property the GRU lacks. Our modified streaming MISRGRU reaches the batch baseline by the twentieth accumulated frame and keeps improving. Augmenting this model with a deformable phase-space alignment module and training it with a motion-augmentation pipeline that simulates moving structures further sustains reconstruction quality under inter-frame sample motion up to speeds of $5\text{ }\mu\text{m}$ per second. By emitting one image per incoming frame instead of reprocessing the whole window, the streaming model with a deformable phase-space alignment cuts the per-output latency from 170 ms to 17 ms, a $10\times$ speed-up over the original RESURF work. Together these results bring deep-learning super-resolution microscopy closer to real-time use.

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Nomenclature

Abbreviations

Abbreviation	Definition
CNN	Convolutional Neural Network
ConvGRU	Convolutional Gated Recurrent Unit
DPA	Deformable Phase-space Alignment
DPA-TISR	Deformable Phase-space Alignment for Time-lapse Image Super-Resolution
DSAIT	Data Science and Artificial Intelligence Technology
FH	Fast-blinking, High-SNR condition
FL	Fast-blinking, Low-SNR condition
FP16	16-bit Floating-Point precision
FPS	Frames Per Second
GAP	Global Average Pooling
GPU	Graphics Processing Unit
GRU	Gated Recurrent Unit
GT	Ground Truth
HPC	High-Performance Computing
HR	High-Resolution
INT8	8-bit Integer precision
LR	Low-Resolution
MISR	Multi-Image Super-Resolution
MISRGRU	Multi-Image Super-Resolution Gated Recurrent Unit
MSA	Multi-Head Self-Attention
MSE	Mean Squared Error
NA	Numerical Aperture
PALM	Photo-Activated Localization Microscopy
PReLU	Parametric Rectified Linear Unit
PSF	Point Spread Function
PSNR	Peak Signal-to-Noise Ratio
RQ	Research Question
RSP	Resolution-Scaled Pearson correlation
SH	Slow-blinking, High-SNR condition
SIM	Structured Illumination Microscopy
SISR	Single-Image Super-Resolution
SL	Slow-blinking, Low-SNR condition
SMLM	Single-Molecule Localization Microscopy
SNR	Signal-to-Noise Ratio
SOFI	Super-Resolution Optical Fluctuation Imaging
SR	Super-Resolution
STED	Stimulated Emission Depletion
TBPTT	Truncated Backpropagation Through Time
TR-MISR	Transformer-based Multi-Image Super-Resolution

Introduction

The living cell is in constant motion. Proteins diffuse, assemble, and disassemble; organelles reshape and migrate; signalling cascades propagate on timescales of seconds to minutes. Capturing these dynamic processes demands simultaneous high spatial and temporal resolution, a combination that conventional optical microscopy cannot provide. The diffraction limit of light restricts spatial resolution to roughly 200–300 nm [5], far larger than the nanometre-scale structures that govern cellular function. These structures are entirely hidden within the diffraction blur. Resolving these structures is important because their shape and arrangement determine how a cell works. To truly understand and combat complex threats like cancer, neurodegenerative disorders, and infections, we must observe how they operate inside living cells.

Fluorescence microscopy addresses this challenge by labelling structures of interest with fluorescent molecules (fluorophores). Upon excitation with a laser, fluorophores emit light that can be detected and imaged. Several super-resolution microscopy techniques have been developed to surpass the diffraction limit and recover sub-diffraction information. However, these methods typically involve significant trade-offs between spatial resolution, acquisition speed, phototoxicity, hardware complexity, and computational cost. These trade-offs are especially acute in live cells, where high excitation intensity and long exposures cause phototoxicity and photobleaching that damage the studied sample. Live-cell imaging is therefore constrained to low excitation power and short exposure times, which limit the number of photons collected per frame. This low photon budget translates directly into images with low signal-to-noise ratio (SNR). Any practical method must be able to resolve the structures at this regime.

Structured Illumination Microscopy (SIM) [15, 14] and Stimulated Emission Depletion (STED) [18] achieve super-resolution through specialised optical hardware and illumination strategies. Single-Molecule Localisation Microscopy (SMLM) methods [4, 31], achieve even higher resolution by localising individual fluorophores. However, SMLM requires sparse stochastic blinking and typically relies on tens of thousands of frames to reconstruct a single super-resolved image. Acquisition times can therefore range from minutes to hours, severely limiting temporal resolution and precluding real-time observation of dynamic cellular processes.

Fluctuation-based methods (Super-Resolution Optical Fluctuation Imaging (SOFI) [9] and Super-Resolution Radial Fluctuations (SRRF) [16]) offer a more favourable balance between spatial and temporal resolution. These methods operate with dense fluorophore labelling and can achieve up to n -fold spatial improvement using higher order statistics.

Despite these advantages, SOFI still requires hundreds of frames per reconstruction. Each SR image is computed offline from the entire acquired sequence, with computational latency in the range of seconds even with GPU acceleration [22]. This is incompatible with real-time live-cell imaging.

The cost of this offline latency is more than inconvenience. When frames are reconstructed only after the acquisition, the experimenter stays blind to the result until it is over. Many failures such as photobleaching, an incorrect exposure, or the cell drifting out of the field of view go unnoticed until

the sample is already wasted. Real-time reconstruction closes this loop: it lets the operator watch the super-resolved structure as it forms and intervene while the experiment runs, refocusing, repositioning, adjusting parameters, or aborting early. Latency is therefore not a secondary optimisation but the property that decides whether super-resolution microscopy can interact with living systems at all.

Deep learning offers a promising route to reduce both the number of frames required and the inference latency. Rather than computing cumulants analytically, a neural network can learn to map a short sequence of low-resolution (LR) fluorescence frames directly to a second-order SR image, effectively approximating the statistical operation from far fewer observations.

Tekpinar et al. [22] introduced SOFI-MISRGRU, adapting the Multi-Image Super-Resolution GRU (MISRGRU) architecture [30] (originally developed for satellite imagery) to the fluorescence microscopy domain. The model demonstrated a 400-fold latency reduction relative to conventional SOFI establishing the practical feasibility of real-time deep-learning SR microscopy.

In this thesis we explore whether the transformer-based fusion module TR-MISR [1] can match or surpass the SR quality of the SOFI-MISRGRU [22] baseline on simulated fluorescence microscopy data across varying SNR conditions.

Like MISRGRU, TR-MISR was originally developed for the PROBA-V challenge [1], a satellite multi-image super-resolution benchmark in which several low-resolution Earth-observation images of the same scene, captured at different revisit times, are fused into a single high-resolution image. On this benchmark TR-MISR outperformed MISRGRU and proved more robust to noisy inputs, a property its authors attribute to the attention mechanism. Whether this advantage transfers to the fluorescence-microscopy setting, where the input is a temporal sequence of blinking frames rather than multiple satellite views, is an open question this thesis addresses.

Secondly, we also extend SOFI-MISRGRU into a continuous-output streaming architecture that eliminates the batch acquisition bottleneck entirely. With this design the model can combine more input images, which enhances the reconstruction quality while enabling uninterrupted super-resolution video generation from dynamic samples in real-time.

1.1. Research Questions

The SOFI-MISRGRU baseline establishes that deep learning can recover second-order SOFI quality from a handful of frames, but it leaves two limitations open. Its ConvGRU fusion is *sequential*, so each frame must wait for the previous hidden state, and it aggregates frames through an unnormalised summation whose magnitude grows with the number of inputs, preventing the model from generalising beyond its training frame count. Secondly, the GAP summates redundant similar information together. Both limitations are addressed in this thesis.

- RQ1.** Does parallel, attention-based fusion (TR-MISR) match or surpass the recurrent ConvGRU fusion (SOFI-MISRGRU) in super-resolution quality across signal-to-noise and blinking conditions, and at what inference-latency cost?
- RQ2.** Can SOFI-MISRGRU be restructured into a streaming architecture that emits one super-resolved image per incoming frame, enabling continuous real-time SR video, and how does its quality and per-output latency compare to the batch model?
- RQ3.** How does inter-frame sample motion degrade SR quality, and does fine-tuning with motion augmentation recover it?

1.2. Contributions

In answering these questions, this thesis makes the following contributions:

1. Adaptation of the TR-MISR transformer fusion module to second-order SOFI super-resolution, including a single-channel encoder, a decoder matched to the fractional SOFI output size, and a CrossAttentionAggregator that separates frame-to-frame communication from final feature aggregation. We also correct the attention-score scaling of the original TR-MISR; this fix is required for the multi-head configurations used in our size ablation, but has no effect on the single-head (1h1d) model we ultimately deploy (Chapter 3).

2. A systematic comparison of TR-MISR and SOFI-MISRGRU across various blinking and SNR conditions, showing that attention-based fusion holds its quality under low-SNR, slow-blinking acquisitions where the GRU degrades, but does *not* reduce inference latency over the RESURF [22] baseline (Chapter 5).
3. An analysis of the frame-count generalisation of both architectures, tracing MISRGRU's degradation to its unnormalised aggregation and showing that TR-MISR trained on eight frames can be evaluated on longer sequences without retraining.
4. A streaming SOFI-MISRGRU that emits one super-resolved output per input frame, reaching the reconstruction quality of batch baseline and improving with more accumulated frames while cutting the per-output latency to RESURF [22] baseline by 10 times. This is the main low-latency contribution of the thesis.
5. A motion-robust extension of the streaming model, combining motion-augmented training with a Deformable Phase-space Alignment module that sustains reconstruction quality under inter-frame sample drift.

1.3. Thesis Outline

Chapter 2 provides background on fluorescence microscopy, SOFI theory, and the deep-learning building blocks underlying both architectures. Chapter 3 describes the adaptations made to TR-MISR and MISRGRU for this application, including the encoder, Transformer fusion module, decoder variants, and the motion augmentation training strategy. Chapter 4 details the synthetic simulation pipeline and training and test set construction. Chapter 5 presents the experimental results, addressing each research question in turn. Chapter 6 interprets the findings in the context of real-time live-cell imaging and identifies limitations.

2

Background

This chapter provides the technical foundation needed to follow the rest of the thesis. It moves from the physics of the imaging problem to the algorithms that this work seeks to accelerate. We begin with fluorescence microscopy and the diffraction limit that motivates super-resolution, then introduce super-resolution techniques and difficulties when implementing them for live-cell imaging. We then cover the deep-learning building blocks used later, and close by describing the two reference architectures this thesis builds on and compares, MISRGRU and TRMISR.

2.1. Fluorescence Microscopy

Fluorescence microscopy enables the visualisation of biological structures at the molecular level by labelling specific targets with fluorescent molecules: *fluorophores*. They emit light upon excitation with a laser. A fluorophore absorbs a photon at the excitation wavelength and re-emits a lower-energy (longer-wavelength) photon. Dichroic filter is used to separate the emitted fluorescence from the excitation light, so only signal from labelled structures reaches the sensor. This way the emitted photons from single molecules can be captured by camera, enabling live-cell imaging of subcellular dynamics.

2.1.1. Diffraction Limit

The spatial resolution of any far-field optical microscope is fundamentally bounded by the wave nature of light. When a point source of light passes through a circular aperture (the objective lens), it produces an *Airy disk* diffraction pattern rather than a perfect point image. Lord Rayleigh formalised this as a resolution criterion: two point sources can just be resolved when the principal maximum of one pattern coincides with the first minimum of the other [5] (Figure 2.1). This gives the minimum resolvable distance:

$$d_{xy} = 0.61 \frac{\lambda}{\text{NA}}, \quad (2.1)$$

where λ is the emission wavelength and $\text{NA} = n \sin \theta$ is the numerical aperture of the objective, determined by the refractive index n of the immersion medium and the half-angle θ of the light cone collected by the lens. For typical visible-light fluorescence ($\lambda \approx 500\text{--}700\text{ nm}$, $\text{NA} \approx 1.4$), this yields $d_{xy} \approx 200\text{--}300\text{ nm}$.

Many biologically relevant structures are far smaller than this limit. Microtubules ($\approx 18\text{ nm}$) and mitochondrial membranes ($\approx 10\text{ nm}$) [3, 32] are entirely hidden within the diffraction blur, making super-resolution a prerequisite for visualising subcellular processes.

2.1.2. Point Spread Function

The *Point Spread Function* (PSF) $v(r)$ describes how a microscope blurs a single point emitter into a diffraction-limited spot. In the focal plane it approximates an Airy disk, which is commonly modelled as a Gaussian $v(r) \propto \exp(-r^2/2\sigma^2)$ [21]. The observed image is a combination of shifted, scaled PSFs

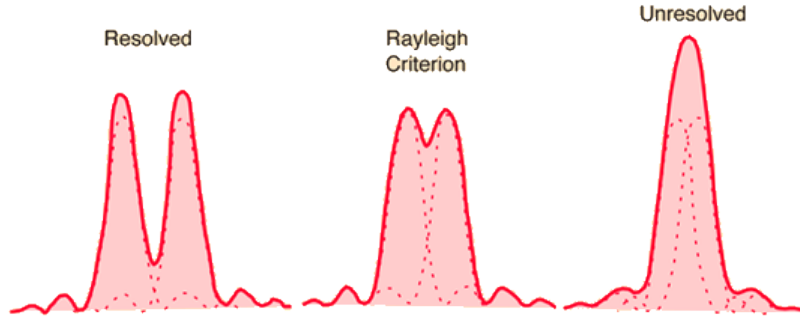


Figure 2.1: The Rayleigh criterion. Two point sources are just resolved when the principal maximum of one Airy pattern coincides with the first minimum of the other.

from every emitter:

$$y(r, t) = \sum_k v(r - r_k) \epsilon_k s_k(t) + n(r, t), \quad (2.2)$$

where emitter k is at position r_k with brightness ϵ_k and time-varying on/off state $s_k(t)$, and $n(r, t)$ is detector noise. Because $v(r)$ is the same at every time step, the temporal variation in $y(r, t)$ comes entirely from the blinking terms $s_k(t)$, on which both various super-resolution techniques rely.

2.1.3. The Deconvolution Problem

Recovering the true structure from the observed blurred image requires inverting the convolution: *deconvolution problem*. This is ill-posed: in the frequency domain, $\hat{Y}(\omega) = \hat{H}(\omega)\hat{X}(\omega) + \hat{N}(\omega)$, the naive inverse $\hat{X} = \hat{Y}/\hat{H}$ amplifies noise at frequencies where $|\hat{H}(\omega)|$ is small, which is near and beyond the diffraction cut-off. Classical regularised methods like Wiener filtering, Lucy-Richardson [26] deconvolution stabilise the inversion but require accurate knowledge of the PSF and yield a single restored image per frame, exploiting no temporal information.

2.1.4. Classical Super-Resolution Microscopy Techniques

Super-resolution (SR) microscopy techniques overcome the diffraction limit by different physical principles.

Structured Illumination Microscopy (SIM) illuminates the sample with a known striped light pattern. The interaction between this pattern and the sample creates an interference pattern (called a moiré fringe) detectable by microscope. Since the illumination pattern is known, the sample structure can be computationally recovered from a series of differently illuminated images. SIM achieves roughly 100 nm lateral resolution; a factor-of-two improvement over the diffraction limit [14, 12]. However, SIM requires specialised and expensive hardware, and slow acquisition of multiple patterned images per reconstruction makes it poorly suited for fast live-cell imaging. High illumination intensity also increases phototoxicity, limiting its use with sensitive biological samples.

Stimulated Emission Depletion (STED) microscopy [18] and Single-Molecule Localization Microscopy (SMLM) can push resolution down to 20 nm. STED narrows the effective PSF by overlaying a doughnut-shaped depletion laser on the excitation spot, suppressing fluorescence at the periphery via stimulated emission. Only fluorophores at the very centre, where depletion intensity is zero, remain able to emit, reducing the emitting volume well below the diffraction limit. Scanning both beams across the sample yields a super-resolved image [19]. Though the STED comes with a cost of long acquisition times, need for specialised fluorophores, and expensive equipment.

Single-Molecule Localization Microscopy (SMLM) techniques such as Photo-Activated Localization Microscopy (PALM) [4] and Stochastic Optical Reconstruction Microscopy (STORM) [31] achieves nanometric spatial resolution down to 10 nm using a widefield microscope. Rather than resolving emitters simultaneously, these techniques build a super-resolved image by stochastically activating

sparse subsets of fluorophores across millions of frames and fitting each isolated emitter to a precise sub-pixel position. While these localization algorithms enable highly accurate molecule detection, SMLM requires: specialized photoswitchable fluorophores, high laser intensities, and the acquisition of an extremely large frame stack over million images. The resulting long acquisition times fundamentally limit temporal resolution and raise the risk of phototoxicity and photobleaching [25].

Limitations Despite their resolution gains, SIM, STED, and SMLM all introduce significant practical complexity compared with conventional widefield microscopy. Each requires specialised hardware beyond a standard widefield setup: structured illumination optics for SIM, a precisely aligned doughnut-shaped depletion beam for STED, and high-power excitation lasers together with photoswitchable fluorophores for SMLM. Computational demands are also substantial. SIM requires multi-image reconstruction for each frame, STED often trades imaging speed for resolution because of point-scanning acquisition, and SMLM must localise large numbers of emitters across thousands to millions of frames. While SIM and some STED implementations can support live-cell imaging, real-time super-resolution remains challenging, particularly for SMLM. In contrast, widefield microscopy is comparatively simple and accessible, typically requiring only standard illumination, an objective lens, and a camera. Super-resolution Optical Fluctuation Imaging exploits exactly this setup to achieve super-resolution without any additional hardware.

Super-resolution Optical Fluctuation Imaging (SOFI) uses the temporal and spatial correlations of independent fluorophore blinking, rather than their precise localisation [9, 28]. Because each emitter independently switches between bright (on) and dark (off) states, the fluorescence intensity at every pixel fluctuates over time. These fluctuations are spatially correlated only for photons from the same emitter: signals from different emitters are statistically independent and their cross-correlation averages to zero. SOFI extracts this structure using higher-order cumulants of the pixel intensity time-traces, which narrows the effective PSF and enhances spatial resolution without requiring individual molecule localisation, offering $\sqrt{n}$ -fold resolution improvement with n^{th} order cummulants of a diffraction limited stack of images [22]. This enhancement can be further improved to n -fold by applying linearization and deconvolution or Fourier reweighting after the cumulant calculation [13]. Longer acquisition times and more images taken are required when calculating higher-order SOFI. This is challenging in practice when sample being exposed to laser for longer is bleached and resulting image has low signal-to-noise ratio.

SOFI can typically achieve temporal resolution on the order of seconds. Reconstruction is generally performed offline after acquisition, and even with GPU acceleration the computational latency remains on the order of seconds per frame, making SOFI not suitable for real-time, long-duration live-cell imaging. [22]

Table 2.1 summarises the main families of non deep learning super resolution techniques.

Table 2.1: Comparison of super-resolution fluorescence microscopy techniques.

Technique	Resolution	Frames	Hardware
STED [18]	< 50 nm	Single	Depletion laser + patterned illum.
SIM [15]	≈ 120 nm	9–15	Patterned illum.
SMLM [31, 4]	10–20 nm	10^5 – 10^6	Widefield
SOFI [9]	$\approx n$ -fold with postprocessing	100–2000	Widefield

2.1.5. Super-Resolution with Deep Learning

Deep learning approaches for image super-resolution replace explicit mathematical models with learnable functions that are trained to map low-resolution frames to high-resolution target.

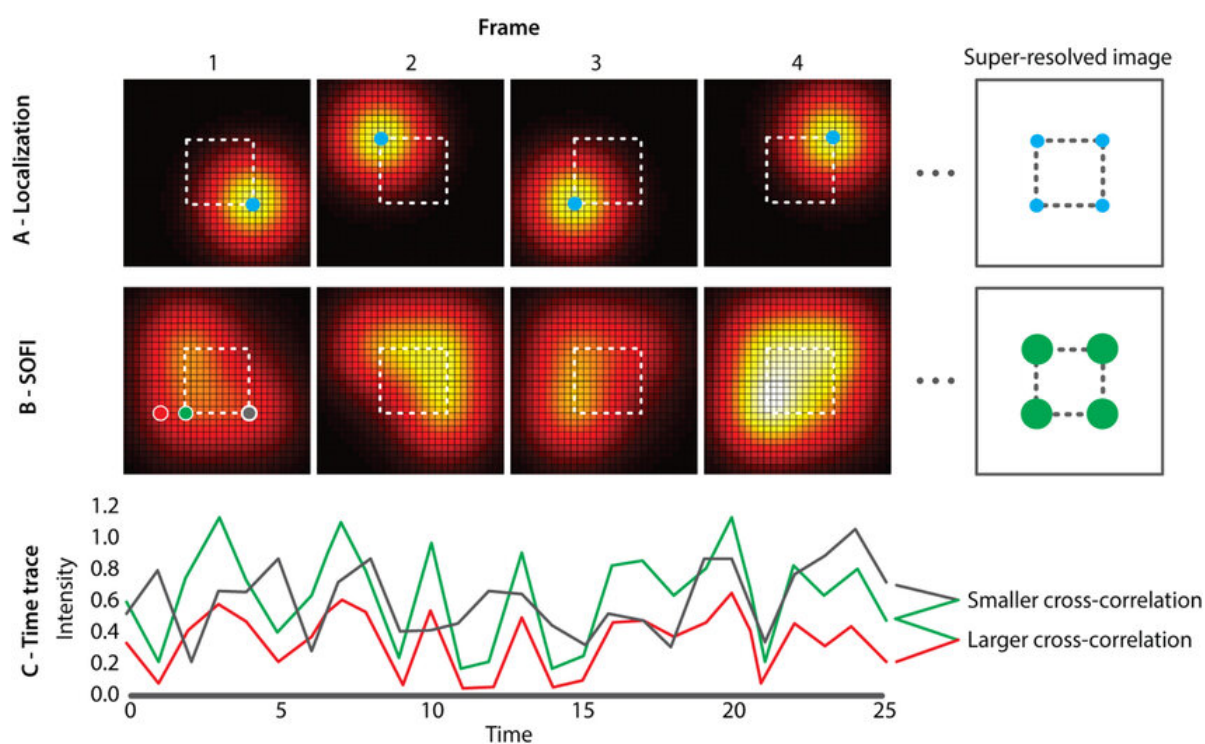


Figure 2.2: Comparison of single-molecule localisation (A) and SOFI (B) on a stack of diffraction-limited frames. Localisation methods pinpoint the centre of each isolated emitter across frames and accumulate the positions into a super-resolved image, requiring sparse, well-separated blinking. SOFI instead exploits the temporal correlations of the intensity time-traces (C): because signals from distinct emitters are statistically independent, their cross-correlation averages to zero, while higher-order cumulants of co-located pixels narrow the effective PSF. This recovers the underlying structure even under dense, overlapping emission where localisation fails.

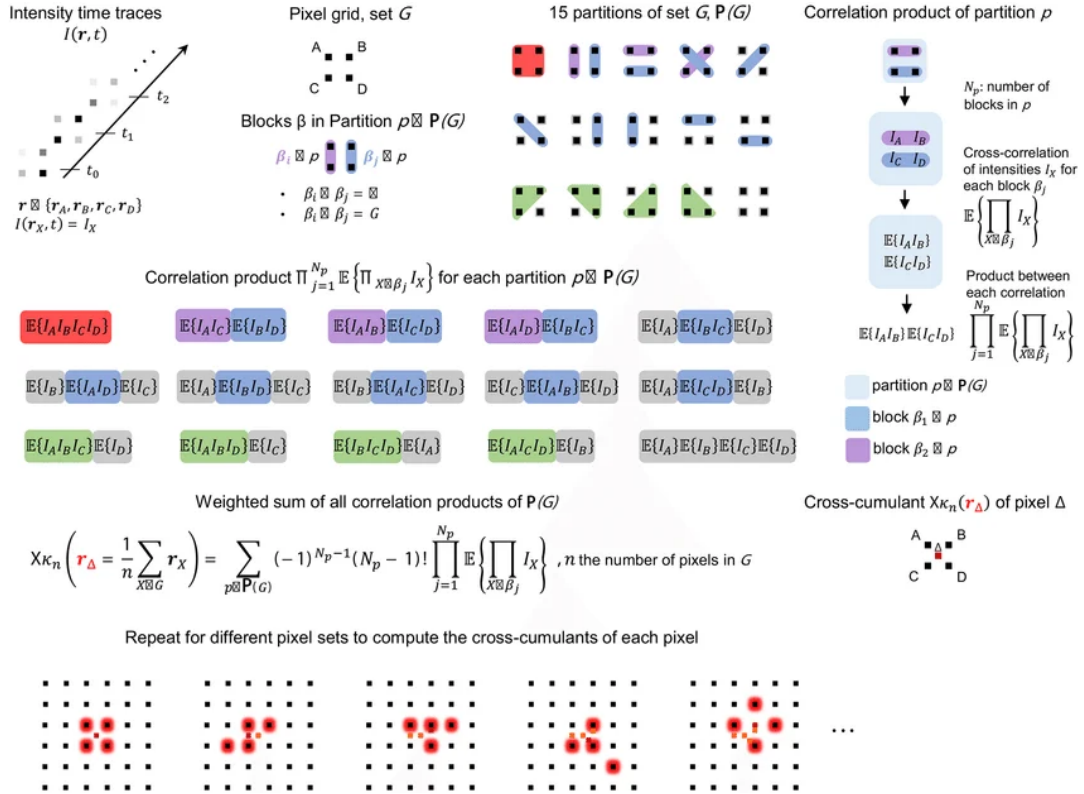


Figure 2.3: SOFI algorithm: cross-cumulant calculation. The n^{th} -order cross-cumulant κ_n of a pixel δ is computed as a weighted sum over all partitions of a set G of n pixels. The position of pixel δ is taken as the geometric mean of the n pixels within G . Choosing different sets of n pixels allows the n^{th} -order cross-cumulant to be evaluated on an arbitrarily large pixel grid. Formulas are given for general n , with sketches for $n = 4$.

Problem Formulation

We define two problem settings: the first is the multi-image super-resolution (MISR) task, which goal is to produce a single high-resolution image from a sequence of low-resolution frames defined as $\{\mathbf{I}_1, \mathbf{I}_2, \dots, \mathbf{I}_N\}$. The second is video super-resolution (VSR), which extends the first by producing a super-resolved image for every frame in the movie, effectively reconstructing a super-resolved video. The high-resolution output is defined as $\mathbf{Y}_t \in \mathbb{R}^{C \times \gamma H \times \gamma W}$ for each time step t , and produced from the sequence of low-resolution frames up to that time step, $\{\mathbf{I}_1, \mathbf{I}_2, \dots, \mathbf{I}_t\} \in \mathbb{R}^{C \times H \times W}$, where C is the number of channels (in our case 1 as the images are grayscale), W and H are the width and height dimensions of low-resolution frames and γ is the upscaling factor.

For second-order SOFI, the high-resolved target has spatial dimensions of $(2H - 7) \times (2W - 7)$. This is due to the nature of SOFI reconstruction algorithm. Specifically, second-order cross-cumulants are computed by pairing pixels within a fixed neighbourhood, with each pair contributing a virtual pixel at its centroid to form a grid twice as dense as the detector array; the limited pairing range means the outermost pixels cannot form valid pairs, which trims the doubled grid and yields the -7 offset. Thus our models need to be designed to produce an output of this specific size, which is not a standard 2 times upscale.

RESURF

Tekpinar et al. [22] proposed the multi-image super-resolution gated recurrent unit (MISRGRU) architecture, inspired by the method used in satellite imaging [30], to achieve real-time fluctuation-based super-resolution (RESURF). The key insight is that the stochastic blinking statistics that SOFI exploits can be recovered from far fewer frames than the classical algorithm requires, provided the fusion network has seen enough representative blinking patterns during training. MISRGRU processes a variable number of low-resolution input frames through a shared CNN encoder followed by a ConvGRU fusion layer, and decodes the resulting hidden state into a single super-resolved output. Crucially, the model is trained end-to-end to approximate the classical SOFI image directly, bypassing the explicit cumulant computation. With as few as 8 frames the model achieves image quality comparable to second-order SOFI, at an inference latency of approximately 27 ms per reconstruction — a roughly 400-fold acceleration over the classical algorithm [22]. Quality is quantified using decorrelation analysis for resolution estimation, and the re-scaled Pearson correlation (RSP) for structural fidelity, commonly used as quality metrics in super-resolution microscopy. The primary limitation of MISRGRU is its sequential fusion stage: because each ConvGRU step depends on the previous hidden state, the fusion cannot be parallelised, and inference time scales linearly with the number of input frames. A detailed architectural description is given in Section 2.2.4.

TR-MISR: Multiimage Super-Resolution Based on Feature Fusion With Transformers

TR-MISR [1] adapts the Transformer to the multi-image super-resolution (MISR) problem, originally demonstrated on the PROBA-V satellite dataset, where N low-resolution images of the same ground patch must be fused into a single high-resolution reconstruction. Rather than processing frames sequentially as a recurrent network does, TR-MISR encodes all N frames in parallel and applies self-attention *across frames* at each spatial location: the N feature vectors at a given pixel form a sequence, and a learnable aggregation token collects cross-frame information via multi-head attention. Because all frames are attended to simultaneously, every frame contributes symmetrically to the output regardless of its position in the input sequence. TR-MISR also demonstrates improved robustness to noisy or partially occluded input frames, a property the authors attribute to the attention mechanism's ability to down-weight uninformative inputs dynamically [1]; ConvGRU-based models lack this capability because their gating is driven by sequential state rather than global frame context. In this thesis we adapt TR-MISR to the fluorescence microscopy super-resolution task, replacing the PixelShuffle decoder with transposed convolutions suited to the non-integer $2n-7$ SOFI upscaling factor. The full architectural description and comparison with MISRGRU are given in Section 2.2.4.

SFSRM: Single-frame deep-learning super-resolution microscopy for intracellular dynamics imaging

Chen et al. [6] proposed single-frame super-resolution microscopy (SFSRM), which takes the opposite approach to fluctuation-based methods: rather than fusing many frames, it reconstructs a super-resolved image from a *single* diffraction-limited frame. The model builds on an enhanced super-resolution

generative adversarial network (ESRGAN), guided by two additions tailored to microscopy data. First, a subpixel edge map is extracted from the input frame and fed to the network as prior information. Second, a multi-component loss combining a multi-scale structural-similarity term with perceptual, adversarial, and frequency-domain losses is used to recover high-frequency detail while suppressing the artefacts that adversarial training can introduce. With these elements SFSRM achieves spatiotemporal resolutions of 30 nm at 10 ms, enabling prolonged live-cell imaging of dynamics such as vesicle transport along microtubules and endosome fusion and fission. However, the method has a prerequisite input signal-to-noise ratio (SNR) of about 15, so under the low-SNR conditions typical of high-speed live-cell imaging it requires a separate signal-enhancement network (a second ESRGAN, denoted SEN) to denoise the frame before reconstruction. This added denoising stage, together with the large size of the adversarial generator, has been reported to result in a computational latency on the order of hundreds of milliseconds in inference time, preventing real-time application [22].

DPA-TISR: Deformable Phase-space Alignment for Time-Lapse SR

Qiao et al. [29] proposed DPA-TISR specifically for long-term live-cell SR imaging using structured illumination microscopy (SIM) data. The model addresses two limitations of single-image SR (SISR) when applied to time-lapse sequences: SISR ignores temporal dependencies between frames, yielding temporally inconsistent reconstructions. SISR provides no calibrated confidence estimate. DPA-TISR combines recurrent propagation with a novel *deformable phase-space alignment* (DPA) module that operates in the Fourier domain: the frequency-domain phase of neighbouring feature maps is aligned before spatial fusion, enabling the model to correct sub-pixel inter-frame motions that optical-flow-based alignment approaches miss. A Bayesian extension (Bayesian DPA-TISR) further produces per-pixel confidence maps, allowing users to identify regions where the SR reconstruction is unreliable. Evaluated on the BioTISR dataset across five biological structures (microtubules, lysosomes, F-actin, mitochondria, clathrin-coated pits), DPA-TISR outperforms both sliding-window and recurrent SISR baselines in PSNR, SSIM, and temporal consistency.

DPA-TISR address related but distinct problem. DPA-TISR targets conventional time-lapse SIM data: the input frames are exploited for their *temporal continuity* (smooth motion between frames), and the model's alignment module is explicitly designed to compensate for that motion. Our work targets the *fluctuation-based* SOFI reconstruction task, where temporal structure is statistically stationary blinking rather than continuous motion, and the goal is to approximate the SOFI cumulant from far fewer frames than the classical algorithm requires. The two settings therefore call for different inductive biases in the temporal fusion module, and results from DPA-TISR do not transfer directly to the SOFI domain.

2.2. Deep Learning for Image Super-Resolution

The following subsections focus on the architectural building blocks most directly relevant to multi-frame SOFI super-resolution.

2.2.1. Convolutional Neural Networks

Convolutional Neural Networks (CNNs) learn local spatial features by convolving an input tensor with a number of learnable filters. Because the same filter is applied at every spatial location, CNNs require far fewer parameters than fully connected networks for image data. In the SR context, CNNs are used as encoders to extract features from frames and decoders to reconstruct the high-resolution output from the representation produced by fusion mechanism (Sections 2.2.2 and 2.2.3).

A key building block in modern SR architectures is the *residual block* [17]. Rather than learning the full mapping $\mathbf{y} = \mathcal{H}(\mathbf{x})$, a residual block learns only the *residual* $\mathcal{F}(\mathbf{x}) = \mathcal{H}(\mathbf{x}) - \mathbf{x}$, so the output is:

$$\mathbf{y} = \mathbf{x} + \mathcal{F}(\mathbf{x}), \quad (2.3)$$

where $\mathcal{F}$ is a sequence of $\text{Conv} \rightarrow \text{PReLU} \rightarrow \text{Conv}$ layers. The identity shortcut allows gradients to flow directly through the block during backpropagation, enabling training of deep networks without vanishing-gradient degradation. In our encoder (Chapter 3), multiple residual blocks process each of the N low-resolution frames independently before the fusion stage.

2.2.2. Recurrent Neural Networks and ConvGRU

Recurrent Neural Networks (RNNs) process sequential data by maintaining a hidden state $\mathbf{h}_t$ that is updated at each step as a function of the current input and the previous state. This allows information from early frames to influence the processing of later frames and in principle enabling the network to aggregate temporal context over long sequences. In practice, vanilla RNNs suffer from vanishing gradients over long sequences.

The *Gated Recurrent Unit* (GRU) [7] addresses this with learned gating mechanisms. Given input $\mathbf{x}_t$ and previous hidden state $\mathbf{h}_{t-1}$, the GRU computes:

$$\mathbf{z}_t = \sigma(\mathbf{W}_z[\mathbf{h}_{t-1}, \mathbf{x}_t]), \quad (2.4)$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r[\mathbf{h}_{t-1}, \mathbf{x}_t]), \quad (2.5)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}[\mathbf{r}_t \odot \mathbf{h}_{t-1}, \mathbf{x}_t]), \quad (2.6)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t, \quad (2.7)$$

where $\mathbf{z}_t$ is the *update gate* (how much of the new frame to incorporate), $\mathbf{r}_t$ is the *reset gate* (how much of the old state to expose to the new frame computation), and $\odot$ denotes element-wise multiplication. The GRU uses fewer parameters than the LSTM while achieving comparable performance on most sequence tasks. [8]

ConvGRU [2] replaces the fully connected weight matrices $\mathbf{W}_z, \mathbf{W}_r, \mathbf{W}$ with 2D convolution kernels, so the hidden state $\mathbf{h}_t \in \mathbb{R}^{C \times H \times W}$ is a spatial feature map rather than a vector. This preserves spatial structure throughout the recurrence, making ConvGRU well suited to image sequence fusion tasks.

2.2.3. Transformers and Attention

The Transformer [35] is a sequence-to-sequence framework built on *self-attention*. Given an input sequence, each element simultaneously queries all other elements to determine how much they relate, producing an output sequence of the same length [1]. Because all elements are processed in a single parallel operation, rather than one step at a time the Transformer avoids the sequential bottleneck of recurrent networks. The cost is that computing the full attention matrix requires $O(N^2)$ operations and memory, which is acceptable for the short sequences ($N \leq 30$) used in this work. The overall structure of the Transformer encoder, with its stacked self-attention and feed-forward blocks, is shown in Figure 2.4.

Scaled Dot-Product Attention

Following the notation of [1], let the input sequence be $X = [x_1, \dots, x_n] \in \mathbb{R}^{d \times n}$ and the output sequence $Y = [y_1, \dots, y_n] \in \mathbb{R}^{d \times n}$. For each $x_i \in X$, the self-attention mechanism applies learned linear projections to obtain a query vector $q_i = W_q x_i \in \mathbb{R}^{d'}$, a key vector $k_i = W_k x_i \in \mathbb{R}^{d'}$, and a value vector $v_i = W_v x_i \in \mathbb{R}^d$, where d' is the projection dimension (which may differ from d). Collecting the key vectors as the matrix $K = [k_1, \dots, k_n] \in \mathbb{R}^{d' \times n}$, the output vector at position i is the attention-weighted sum of all value vectors:

$$y_i = \sum_{j=1}^n \left[\text{softmax} \left(\frac{q_i^\top K}{\sqrt{d'}} \right) \right]_j v_j. \quad (2.8)$$

Each y_i is thus an input-dependent weighted combination of $\{v_j\}_{j=1}^n$; the weights reflect the similarity between query q_i and each key k_j , and are independent of the position of x_i in the sequence [1]. The scaling factor $1/\sqrt{d'}$ prevents the dot products from growing large in magnitude when d' is large, which would push the softmax into regions of very small gradient.

Multi-Head Attention

Rather than performing a single attention operation, multi-head attention [1] maps the query vector q_i and key vector k_i to different subspaces $\{\mu_1 \dots \mu_q\}$ of the representation space and computes attention in each subspace separately, allowing the model to attend to different aspects of the features (Figure 2.5).

Each head $s \in \{1, \dots, q\}$ produces an output sequence $Y^s = [y_1^s, \dots, y_n^s]$ via Eq. (2.8). The q output sequences are concatenated and projected by a learned weight matrix W^o :

$$Z = \text{Concat}(Y^1, \dots, Y^q) W^o, \quad (2.9)$$

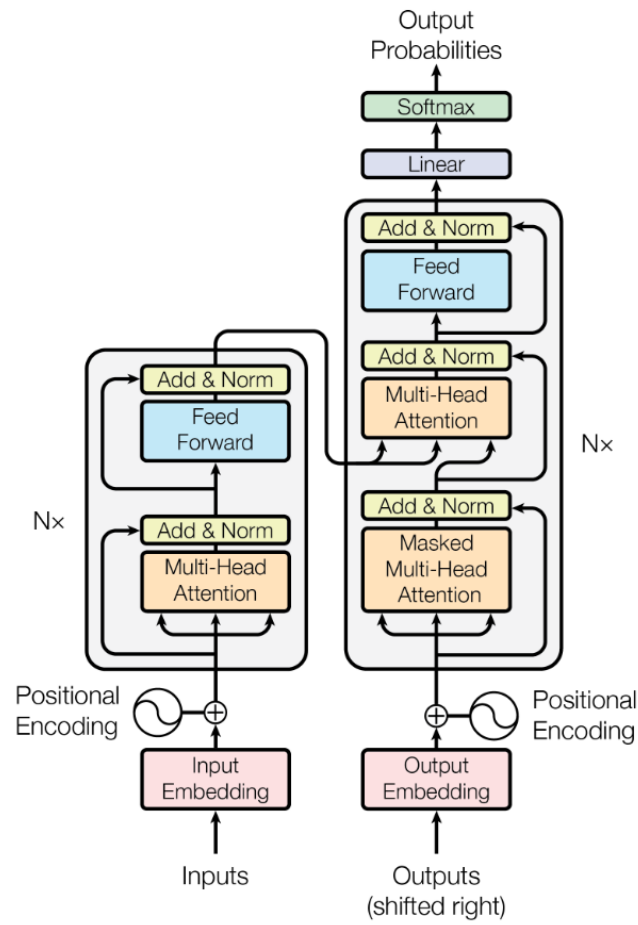


Figure 2.4: The Transformer architecture [35]. Each encoder layer combines multi-head self-attention with a position-wise feed-forward network, using residual connections and layer normalisation.

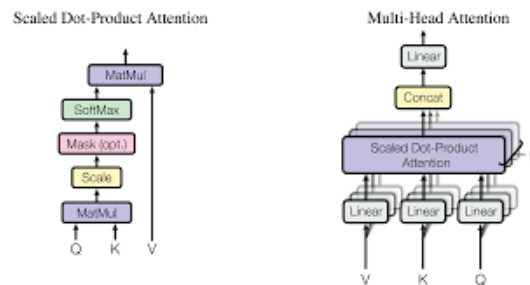


Figure 2.5: Multi-head attention [35]. The queries, keys, and values are linearly projected into subspaces, scaled dot-product attention is computed in each head in parallel, and the head outputs are concatenated and projected back to the input dimension.

where $Z = [z_1, \dots, z_n] \in \mathbb{R}^{d \times n}$ has the same size as the input X . Using $\text{MSA}(\cdot)$ to denote multi-head self-attention, the full operation is:

$$[z_1, \dots, z_n] = \text{MSA}([x_1, \dots, x_n]). \quad (2.10)$$

2.2.4. Multi-Frame Super-Resolution Architectures

The choice of a multi-frame architecture follows directly from the nature of the SOFI task. The point spread function is identical at every time step, so a single frame carries no super-resolution information. It is simply a diffraction-limited image. The resolution gain that SOFI exploits lives entirely in the temporal statistics of fluorophore blinking. A network that processes frames jointly can learn to recover the same statistical information that the classical SOFI algorithm computes explicitly. This is why both reference architectures fuse a sequence of N frames, and it reframes the central design question as how those frames should be fused, the contrast between sequential recurrent fusion and parallel attention-based fusion explored in this thesis.

The contributions in Chapter 3 build directly on two existing multi-frame SR architectures. We describe the original models here so the reader can see what is newly implemented in this thesis.

MISRGRU

Arefin et al. [30] adapted ConvGRU to the *multi-image super-resolution* (MISR) problem for the PROBA-V satellite dataset, where N low-resolution images of the same ground patch must be fused into a single high-resolution reconstruction. The RESURF paper [22] subsequently adapted the architecture to the SOFI fluorescence microscopy domain, demonstrating that it can recover second-order SOFI quality from just 8–20 frames at real-time inference speeds (≈ 27 ms per reconstruction). The architecture consists of three stages: a shared CNN encoder, a ConvGRU fusion layer, and a decoder.

Encoder. Each low-resolution frame $I_t \in \mathbb{R}^{1 \times H \times W}$ is encoded independently by a shared CNN into a feature map $f_t \in \mathbb{R}^{C \times H \times W}$. The encoder uses residual blocks with PReLU activations; all frames share the same encoder weights, so the N frames can be encoded in parallel.

ConvGRU Fusion. The encoded features are processed sequentially through a ConvGRU (Section 2.2.2), accumulating information across frames into a hidden state $\mathbf{h}_t$. After all N frames are processed, the GAP layer summates all hidden states $\mathbf{h}_1 \dots \mathbf{h}_N$ and that is passed to the decoder. The overall architecture is illustrated in Figure 2.6.

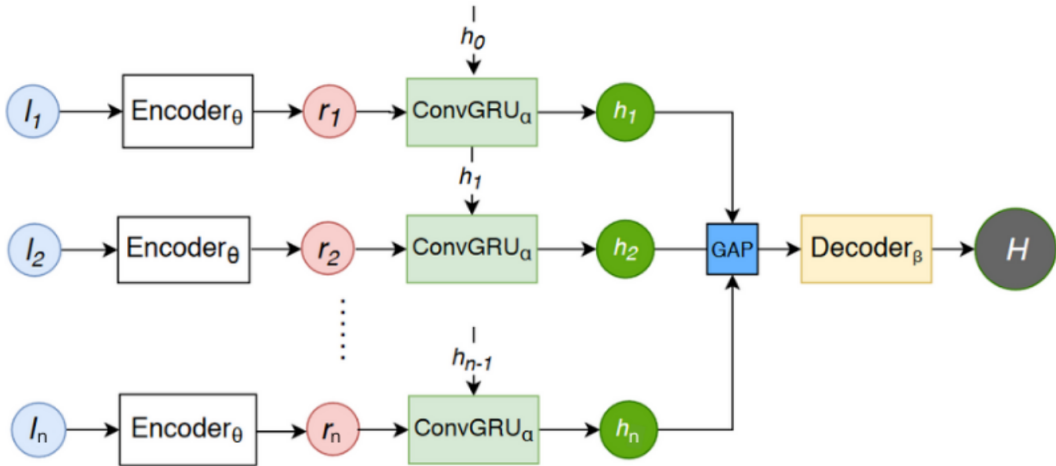


Figure 2.6: MISRGRU architecture. Each LR frame is encoded independently by a shared CNN; the ConvGRU fuses the resulting feature maps sequentially into a single hidden state, which the decoder upsamples to the HR output.

Decoder. A transposed convolution upsamples $\mathbf{h}_N$ to the target size $(2H - 7) \times (2W - 7)$ directly, without any post-hoc crop.

Limitations for SOFI. We identified the following few limitations that we tackle in this thesis: First, *sequential fusion*: while the encoder runs in parallel across frames, the ConvGRU processes one frame at a time each step depends on the previous hidden state, so the fusion stage cannot be parallelised. Inference time in the ConvGRU therefore scales linearly with the number of input frames, even when all frames are available simultaneously. Second, *asymmetric frame weighting*: by the time the GRU reaches frame N , the update gate may have substantially overwritten contributions from early frames. For SOFI, blinking events in frame 1 carry the same statistical weight as those in frame N ; the recurrent bias may systematically underweight early-frame information. Third, *n-frame training limitation*: the model trained on N frames performs best when N frames are used in inference, despite providing more frames in inference time. These limitations motivate exploring a parallel, attention-based fusion module that treats all N frames symmetrically as described in Section 2.2.4.

TR-MISR

Rather than treating 2D spatial patches as tokens, TR-MISR encodes each of the N input frames into a feature map and then applies the Transformer per spatial position: the N feature vectors at each pixel location form a length- N sequence, and a learnable embedding token is prepended to aggregate cross-frame information at that location [1].

Encoder. To help the network align frames implicitly, a reference image is first computed as the pixel-wise median across all input frames [1]:

$$LR_{\text{ref}} = \text{Median}(LR_1, \dots, LR_k). \quad (2.11)$$

Each frame LR_i is concatenated with LR_{ref} to form a 2-channel input, which is then encoded by a shared residual-block CNN to a feature map $f_i \in \mathbb{R}^{H \times W \times C_h}$.

Fusion module. The fusion module keeps the spatial resolution unchanged and applies the Transformer *per spatial location* [1]. For each pixel position (h, w) , the k channel vectors $\{x_{(h,w)}^i\}_{i=1}^k$ from the encoded feature maps form a length- k sequence. A randomly-initialised learnable embedding vector $x_{(h,w)}^0 \in \mathbb{R}^{d_0}$ is prepended to this sequence, giving:

$$z^0 = [x^0, x^1, \dots, x^k]. \quad (2.12)$$

A Boolean mask $\alpha = [1, 1, \dots, 0]$ marks zero-padded placeholder frames so that padded positions are ignored during attention. The sequence passes through M Transformer encoder layers, each consisting of masked multi-head self-attention (MSA) and a feed-forward network (FFN) with pre-LayerNorm and residual connections:

$$z_a^l = \text{MSA}\left(\text{LN}(z^{l-1}, \alpha)\right) + z^{l-1}, \quad (2.13)$$

$$z^l = \text{FFN}\left(\text{LN}(z_a^l)\right) + z_a^l. \quad (2.14)$$

After M layers, the embedding vector at index 0 of the output serves as the fused feature for that pixel position [1]. Assembling z^u across all $H \times W$ positions gives the fused feature map f_u , which is passed to the decoder.

Decoder. The decoder applies a 1×1 convolution to rearrange channels, followed by a PixelShuffle (sub-pixel convolution) with upscale factor $r = 3$ to produce the high-resolution output.

Key properties. Because the Transformer attends to all N frames simultaneously, TR-MISR processes frames in parallel which avoids sequential processing of the ConvGRU. The fusion is also *permutation invariant*: the self-attention treats the N frame features as an unordered set, since no positional encoding is applied along the frame dimension. This aligns naturally with the SOFI algorithm, which estimates cumulants from fluorescence fluctuations and is invariant to the ordering of frames. Therefore, treating the input sequence as an unordered set.

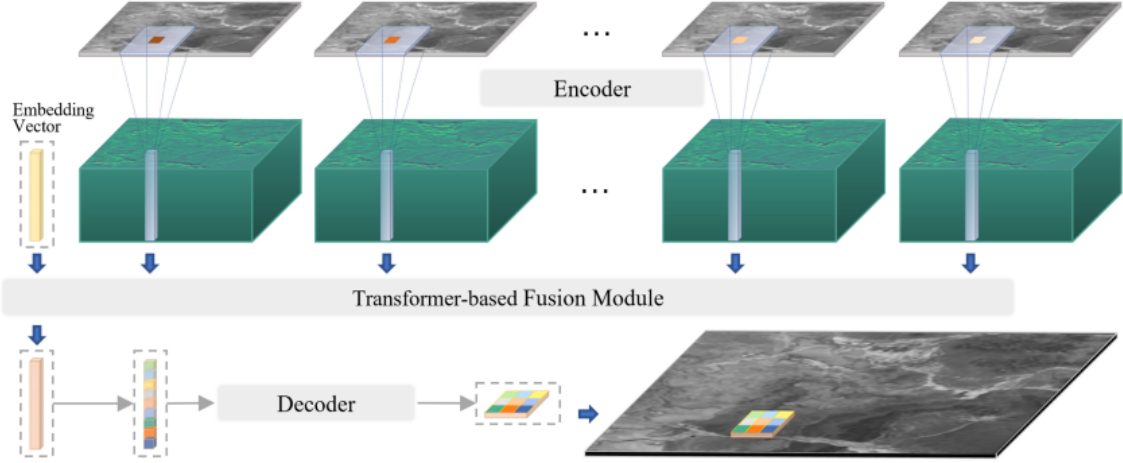


Figure 2.7: Overview of TR-MISR [1]. The encoder encodes the same area of low-resolution images. The transformer based fusion module is responsible for a featurewise fusion by an additional learnable embedding vector. The pixelshuffle decoder decodes all fused features into a high-resolution image.

2.2.5. Loss Functions for Super-Resolution

The choice of training loss shapes what the network learns to optimise; pixel-wise losses are among the most widely used for image SR [36].

Pixel-wise losses. The ℓ_1 (mean absolute error) loss $\mathcal{L}_{L1} = \frac{1}{HW} \|\hat{x} - y\|_1$ and the ℓ_2 (mean squared error) loss $\mathcal{L}_{L2} = \frac{1}{HW} \|\hat{x} - y\|_2^2$ penalise deviations between the prediction $\hat{x}$ and the target y in the spatial domain. ℓ_2 penalises large errors quadratically, which biases the network towards producing blurry, averaged outputs; ℓ_1 is more robust to outliers and tends to preserve sharper edges. Both losses treat all pixel locations equally, which is problematic when the image is dominated by a dark background (as is common in fluorescence microscopy): the loss is then driven primarily by background pixels that carry no SR information.

Fourier-domain loss. We adopt the Fourier-domain loss proposed by [11], penalising errors separately in amplitude and phase:

$$\mathcal{L}_{\mathcal{F}} = \mathcal{L}_{\mathcal{F}A} + \lambda_{\phi} \mathcal{L}_{\mathcal{F}\angle}, \quad (2.15)$$

$$\mathcal{L}_{\mathcal{F}A} = \frac{1}{|\Omega|} \sum_{(u,v) \in \Omega} \left| |\hat{X}_{u,v}| - |Y_{u,v}| \right|, \quad (2.16)$$

$$\mathcal{L}_{\mathcal{F}\angle} = \frac{1}{|\Omega|} \sum_{(u,v) \in \Omega} \left| \text{wrap}(\angle \hat{X}_{u,v} - \angle Y_{u,v}) \right|, \quad (2.17)$$

where $\hat{X} = \mathcal{F}_{\text{ortho}} \hat{x}$ and $Y = \mathcal{F}_{\text{ortho}} y$ are the orthonormal DFTs of the prediction and target, Ω is the set of unique frequency bins retained after exploiting Hermitian symmetry ($|\Omega| = H \times (W/2 + 1)$), and $\lambda_{\phi} = 0.1$ downweights the phase term relative to the amplitude term. The orthonormal transform scales each coefficient by $1/\sqrt{HW}$, making the loss value independent of image resolution.

The amplitude and phase of each coefficient are

$$|X_{u,v}| = \sqrt{\mathcal{R}\{X_{u,v}\}^2 + \mathcal{I}\{X_{u,v}\}^2}, \quad \angle X_{u,v} = \text{atan2}(\mathcal{I}\{X_{u,v}\}, \mathcal{R}\{X_{u,v}\}), \quad (2.18)$$

where atan2 is the two-argument arctangent returning values in $(-\pi, \pi]$.

Phase differences must be wrapped to $(-\pi, \pi]$ to avoid penalising equivalent phases as large errors (a difference of 2π is the same physical phase). The wrapping operator is

$$\text{wrap}(\delta) = ((\delta + \pi) \bmod 2\pi) - \pi, \quad (2.19)$$

which maps any angle difference δ to the shortest circular distance in $(-\pi, \pi]$.

The coefficients are obtained via the 2D real-valued FFT:

$$X_{u,v} = \frac{1}{\sqrt{HW}} \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} x_{h,w} e^{-i2\pi(\frac{uh}{H} + \frac{vw}{W})}, \quad (2.20)$$

where the $1/\sqrt{HW}$ prefactor reflects the orthonormal convention.

Following the RESURF paper [22], which demonstrated strong reconstruction quality with fourier domain loss, we train our models with the fourier loss and a combined ℓ_1 +Fourier loss. Our primary focus is on comparing the Transformer and GRU fusion modules, so we adopt these two loss functions from the paper.

2.2.6. Evaluation Metrics

Standard pixel-level metrics such as MSE, PSNR, and SSIM are poorly suited for evaluating super-resolution SOFI reconstructions. As noted by [21], SOFI images are dominated by near-zero background pixels, so averaging errors over all pixels yields an overly optimistic assessment even when fine structural detail is poorly reproduced. We instead adopt two metrics used by [22], enabling direct comparability.

Resolution-scaled Pearson correlation (RSP). RSP measures the Pearson correlation between a predicted SR image and a reference image, after the SR image has been rescaled in both resolution and intensity to match the reference. The rescaling corrects for differences in sharpness between prediction and reference, so RSP reflects structural agreement rather than penalising the model for being sharper than the reference. RSP lies in $[-1, 1]$; values closer to 1 indicate better structural fidelity. When the ground-truth SOFI image is available (simulated test set), it is used as the reference.

Decorrelation analysis [10] is commonly used in SR microscopy to estimate spatial resolution directly from each output image without requiring a ground truth. The normalised Fourier transform is cross-correlated with the input image through a binary circular mask of shrinking radius $r \in [0, 1]$. A peak in the resulting curve $d(r)$ identifies the highest spatial frequency carrying coherent signal. Repeating over a series of high-pass filters and taking the maximum peak position k_c gives the cut-off frequency, from which resolution is computed as:

$$\text{Resolution} = \frac{2 \times \text{pixel size}}{k_c}. \quad (2.21)$$

This parameter-free estimate yields the achieved spatial resolution in nanometres, complementing the structural-fidelity measure provided by RSP.

RSP is the primary metric reported in all experiments; decorrelation analysis is reported only for the best models of each class.

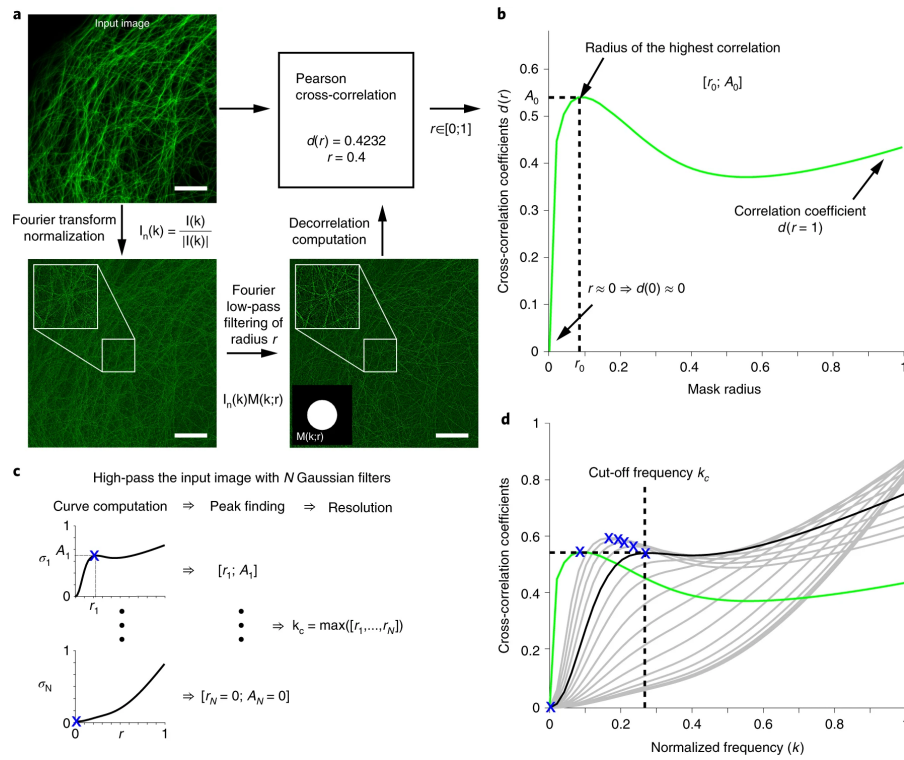


Figure 2.8: a) Cross-correlation of the image with its normalized version after applying a Fourier filter. b) Cross-correlation coefficient plotted as a function of the mask radius. c) High-pass filtering applied to the input image, followed by resolution estimation. d) Plot showing all the decorrelation functions calculated for the image along with resolution estimation. The green line represents the decorrelation function without high-pass filtering, while the grey lines correspond to those with high-pass filtering. The blue crosses indicate local maxima, and the black line highlights the decorrelation function at the highest frequency peak. The vertical dashed line marks the cut-off frequency k_c . Scale bar: 5 μm .

3

Methods

This chapter describes our contributions to this thesis. Background described the base architectures and components of MISRGRU and TR-MISR in Sections 2.2.4 and 2.2.4.

Here we describe only what is new: the modifications made when adapting model architectures to fluorescence microscopy, an improved Transformer fusion module, a streaming variant of MISRGRU, and a motion augmentation training strategy.

3.1. Modifications to TRMISR

Encoder. Our modified TRMISR shares the same encoder design as RESURF [22]. The original TR-MISR encoder takes a two-channel input formed by concatenating each grayscale low-resolution frame with a median reference image. In TR-MISR the median reference serves as an implicit co-registration anchor and a shared representation for separating redundant content from frame-unique detail [1].

We drop the median reference and use a single-channel input. Removing the median does not discard the information that drives super-resolution. The resolution gain in SOFI comes from the blinking-induced intensity fluctuations across the frame ensemble.

This is tested out empirically: an ablation that concatenates an averaged reference image as a second encoder channel is significantly worse than the single-channel model across every imaging condition and on every paired test sample ($\bar{d} = +0.027$ RSP pooled in favour of the single channel, $p \approx 2.3 \times 10^{-80}$, rank-biserial = -1.00), with the gap widening as the blinking slows and the photon count drops (Appendix B.1).

We reduce the encoder width from 64 to 24 channels. Reducing the encoder width from 64 to 24 channels roughly halves inference latency at every frame count (e.g. 226 ms vs. 436 ms at 20 frames for 512x512 input images), while costing only a small drop in reconstruction quality (aggregate RSP 0.8925 vs. 0.8955, a statistically significant but minor 0.003 decrease that widens to 0.006 in the hardest slow-blinking, low-SNR condition). We therefore use the 24-channel encoder for all fusion-mechanism comparisons, both to match the RESURF configuration [22] and because the latency saving outweighs the marginal quality loss; the wider 64-channel encoder is preferable when quality is prioritised.

Decoder. The original TR-MISR decoder uses a $3\times$ PixelShuffle [33, 37] for the $3\times$ upscaling. PixelShuffle (sub-pixel convolution) rearranges the channels of a low-resolution feature map of shape $(C \cdot r^2, H, W)$ into a high-resolution map of shape (C, rH, rW) , performing the upscaling by an integer factor r without learnable upsampling parameters (Figure 3.1). For the $(2H - 7) \times (2W - 7)$ SOFI upscale, we compare two decoder designs: a $2\times$ PixelShuffle followed by a centre-crop to match the target resolution, and a learned transposed convolution that achieves the exact $(2H - 7) \times (2W - 7)$ output size directly. These are described in detail in Section 3.3.

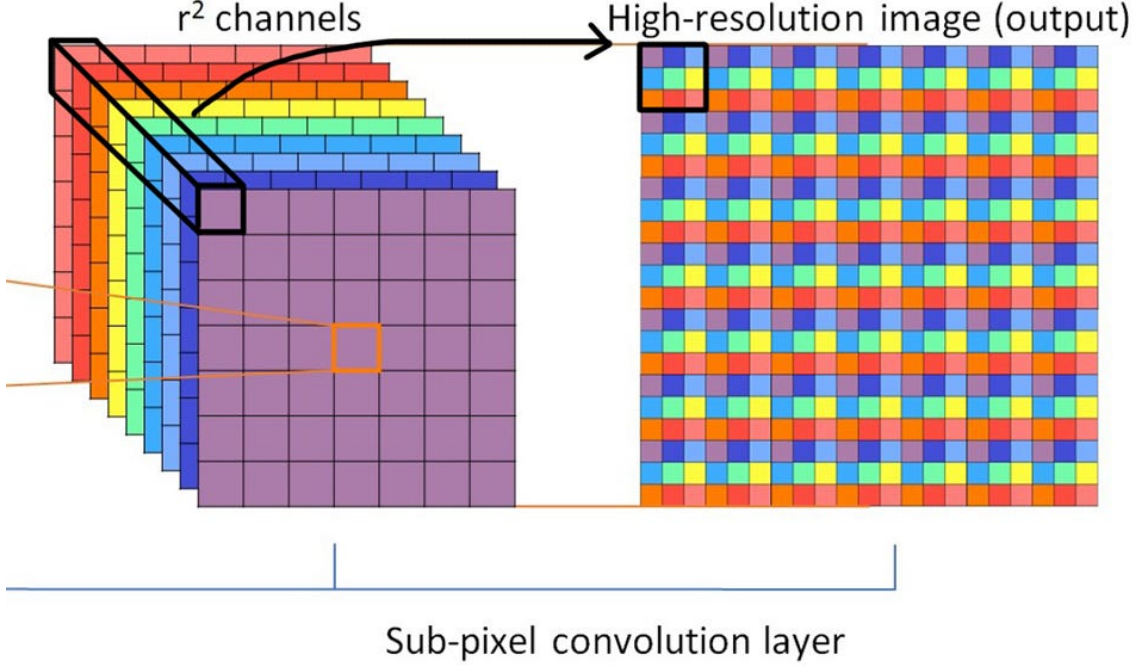


Figure 3.1: PixelShuffle (sub-pixel convolution) upscaling by a factor r : the $(C \cdot r^2, H, W)$ feature map is rearranged into a (C, rH, rW) high-resolution output [33, 37].

3.2. Improved Fusion

The base TR-MISR fusion module is described in Section 2.2.4, but in short it pays attention to encoded input frames by fusing the information into an additional learnable embedding vector. For every pixel (h, w) , in the sequence of N frames: the embedding token x^0 is prepended (Eq. (2.12)), and after M Transformer layers the token at index 0 is extracted as the fused feature (Eq. (2.13)–(2.14)). We identified two problems with this design and propose fixes for both.

3.2.1. Problem 1: Attention Scaling Bug

The original TR-MISR code scales attention scores by $1/\sqrt{d_{\text{model}}}$, where d_{model} is the full embedding dimension. The standard [35] is to scale by $1/\sqrt{d_k}$, where $d_k = d_{\text{model}}/q$ is the per-head dimension. The two differ by a factor of $\sqrt{q}$, meaning attention logits are under-scaled, which pushes the softmax towards a uniform distribution, every frame receives roughly equal weight regardless of its signal quality.

We replace the manual attention implementation with `nn.MultiheadAttention`¹, which applies the correct $1/\sqrt{d_k}$ scaling internally. However, this bug only affects multi-headed attention. In the final TRMISR design we use single head, and single depth attention fusion mechanism, since our ablation study shows that increasing the number of heads and attention depth increases latency without significantly improving accuracy.

3.2.2. Problem 2: Class Token replaced with CrossAttentionAggregator

In the original TR-MISR, a single learnable class token is added into the self-attention sequence together with the N frame tokens. This token is expected to do two things at once: help the frame tokens exchange information with each other, and collect the final fused representation.

CrossAttentionAggregator. We separate the two roles. First, the N frame tokens attend to each other through self-attention layers without the class token, thus all information is used for frame-to-frame communication. Then a single learned query vector $\mathbf{Q}$ cross-attends to the updated frame tokens to

¹<https://docs.pytorch.org/docs/stable/generated/torch.nn.MultiheadAttention.html>

produce the fused feature:

$$\mathbf{z}'' = \text{CrossAttn}(\mathbf{Q}, \{x^1, \dots, x^k\}). \quad (3.1)$$

The full forward pass is therefore: (1) add positional embeddings to the frame tokens; (2) run M self-attention layers on the frame tokens only; (3) aggregate with the cross-attention query to obtain $\mathbf{z}''$; (4) decode $\mathbf{z}''$ to the SR output. The CrossAttentionAggregator is similar to attention-based pooling (PMA) in Set Transformer[23], which has been shown to outperform fixed pooling strategies for set representation learning tasks.

3.3. PixelShuffle vs Convolutional decoder

The decoder receives a fused spatial feature map $\mathbf{z} \in \mathbb{R}^{C \times H \times W}$ and must produce the target SOFI image $\hat{\mathbf{Y}} \in \mathbb{R}^{1 \times (2H-7) \times (2W-7)}$. Two decoder designs were compared to handle this non-integer upscaling.

PixelShuffle decoder (TR-MISR baseline). PixelShuffle [33], is an efficient and parameter-free spatial upsampler [1]. It rearranges feature channels from a low-resolution feature map into higher-resolution output. Moreover, this approach avoids the checkerboard artefacts that transposed convolutions can introduce when their stride does not divide evenly into the kernel size [27]. More specifically, let the input feature map have dimensions $H \times W \times C_{\text{in}} r^2$, where r is the upscaling factor. The pixel shuffle operation transforms this tensor into an output of size $rH \times rW \times C_{\text{in}}$ by redistributing the r^2 channels associated to each spatial location into an $r \times r$ spatial neighborhood in the high-resolution image.

For the PROBA-V satellite benchmark, the target resolution was an exact $3\times$ upscale of the input. We modified the original network for our use case, so a $2\times$ PixelShuffle is applied. Our decoder consists of the following steps:

1. A 3×3 convolutional layer with PReLU activation decreases the number of channels from 24 to $4C$.
2. PixelShuffle rearranges the $4 \times H \times W$ tensor into $1 \times 2H \times 2W$.
3. A centre-crop operation removes 3 pixels from the top/left and 4 pixels from the bottom/right, which results in the desired output size.

The crop introduces a asymmetric offset in the predicted image and does not back-propagate any learnable upsampling signal through the cropped region. This is a known limitation of PixelShuffle when the target size is not an exact power-of-two multiple of the input.

ConvTranspose2d decoder (TRNet_deconv and MISRGRU). The original MISRGRU [30] and RESURF [22] models use a transposed convolutional layer to upsample the fused features. The key advantage over PixelShuffle is that the upsampling kernel is fully learned and the $2H - 7$ target size is achieved without any explicit crop, so the decoder's receptive field covers all output pixels:

$$\hat{\mathbf{Y}} = \mathcal{D}_\beta(\mathbf{z}) \in \mathbb{R}^{1 \times (2H-7) \times (2W-7)}. \quad (3.2)$$

The decoder consists of the following layers:

1. ConvTranspose2d($C \rightarrow C$, kernel 3×3 , stride 2, padding 4, output_padding 0)
2. Two 3×3 convolutional layers with PReLU refine the upsampled features.
3. A 1×1 convolutional projection reduces from C channels to a single output channel.

All three layers maintain $C = 24$ channels until the final projection.

The output spatial dimension of a transposed convolution with kernel size k , stride s , padding p , and output padding p_{out} is:

$$H_{\text{out}} = (H_{\text{in}} - 1) \cdot s - 2p + k + p_{\text{out}}. \quad (3.3)$$

Substituting the decoder parameters ($s=2$, $p=4$, $k=3$, $p_{\text{out}}=0$) gives:

$$H_{\text{out}} = (H - 1) \cdot 2 - 2 \cdot 4 + 3 = 2H - 7, \quad (3.4)$$

which matches the 2nd-order SOFI target exactly.

3.4. Optimizing MISRGRU

The original MISRGRU model (Section 2.2.4) takes a fixed set of N input frames and outputs a single SR image. This has two limitations for real-time live-cell imaging. First, for every new incoming frame the entire N -frame sequence must be re-encoded and re-processed from scratch, which is redundant since the ConvGRU hidden state $\mathbf{h}_N$ already encodes a summary of all previous frames. Second, the batch model is many-to-one: it produces one output per N input frames. To super-resolve a full movie, the entire sequence would need to be processed in rolling windows which again introduces the computational redundancy.

This section develops a family of MISRGRU variants that progressively address these limitations. Table 3.1 previews all six models, their architectural modifications, and how they are trained; each is described in detail in the subsections that follow and evaluated in Chapter 5.

Table 3.1: Overview of the MISRGRU model variants studied in this thesis. The output mode $N \rightarrow 1$ denotes one SR image per N input frames (batch); $N \rightarrow N$ denotes one SR output per incoming frame (streaming). All streaming variants reuse the encoder and decoder of the batch model and differ only in hidden-state handling, added modules, and training.

Model	Output	Modification	Training strategy
Batch MISRGRU (foundational)	$N \rightarrow 1$	RESURF[22] ConvGRU baseline; GAP aggregates all hidden states $h_1 \dots h_N$; state reset per sample.	Fourier loss on model output and SOFI target.
Sliding-window MISRGRU	$N \rightarrow N$	Circular buffer of the last n hidden states; hidden states are cached only n -th is computed. GAP aggregates only the windowed states before decoding.	No training needed, reuses weights from the Batch MISRGRU.
Streaming MISRGRU	$N \rightarrow N$	GAP module removed; decoder reconstructs from the last hidden state h_k alone; state never reset. Ability to aggregate more information over time.	Truncated BPTT (TBPTT) with chunking and <code>loss_last_n</code> supervision; Fourier loss.
Larger GRU	$N \rightarrow N$	Streaming MISRGRU with increased ConvGRU hidden-state capacity (95k $\rightarrow$ 131k params).	Trained from scratch with TBPTT (≈ 190 epochs, DelftBlue); gives best static accuracy.
Fine-tuned MISRGRU	$N \rightarrow N$	Streaming MISRGRU architecture, no new modules.	Initialised from baseline Fourier weights and fine-tuned for 100 epochs on motion-augmented data.
MISRGRU with DPA	$N \rightarrow N$	Adds a Deformable Phase-space Alignment module (RFFT phase correction + deformable convolution) that aligns h_{t-1} to f_t before the recurrent update.	Zero-initialised module fine-tuned from baseline Fourier weights for 300 epochs on motion-augmented data.

3.4.1. Converting MISRGRU to many-to-many

An architecture where many inputs produce many outputs would address both problems: one SR image is produced per incoming frame, and the GRU hidden state is carried across frames rather than reset between batches. This way the model could accumulate more information from input frames, which can help to better reconstruct, while keeping the latency low per reconstructed image.

Sliding-window MISRGRU

A circular buffer of the last n hidden states is maintained across time steps. At each step k , the encoder processes the incoming frame $\mathbf{I}_t$, the ConvGRU updates its state, and the GAP module aggregates features from the last n hidden states before decoding to produce $\hat{\mathbf{Y}}_t$. This design retains the multi-frame fusion capability of batch MISRGRU while enabling streaming inference without resetting the hidden state every K frames, while also saving compute by remembering $h_{k-n} \dots h_{k-1}$.

We designed experiments in section 5.2.1 to see how this architecture without reinitializing the hidden state performs when run longer than 20 frames. The results from the experiments prompted us to remove the GAP layer and force the Decoder to reconstruct the SR output only from last hidden state h_k .

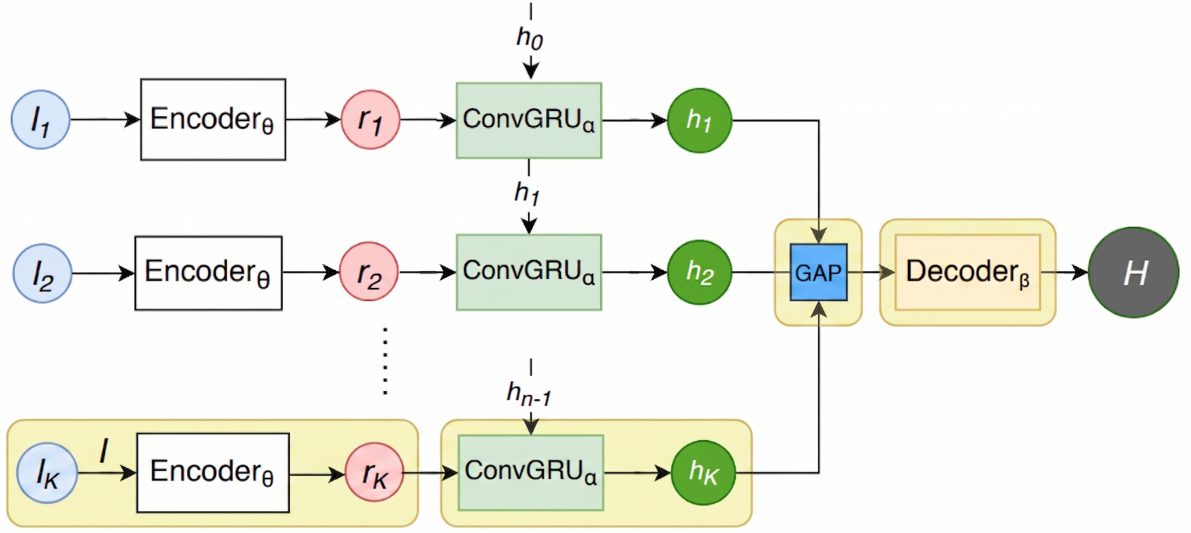


Figure 3.2: Sliding-window MISRGRU architecture. Only highlighted parts are being computed at arrival of new input frame. ConvGRU outputs $h_{k-n} \dots h_{k-1}$ are being stored in circular buffer for faster inference

Removing the GAP module

The Streaming MISRGRU (Figure 3.3) reuses the encoder and decoder from the batch MISRGRU without modification. The only architectural change is in how the ConvGRU hidden state is managed. In the batch model, the hidden state is zeroed at the start of each new sample. In the streaming model, h_t is carried across successive frames and never reset between time steps, so the GRU accumulates context from the entire history of frames seen so far. The GAP in RESURF[22] is computed as sum of all $h_1 \dots h_k$. Removing it forces both ConvGRU and Decoder to reconstruct the SR output from last hidden state h_k at every new frame.

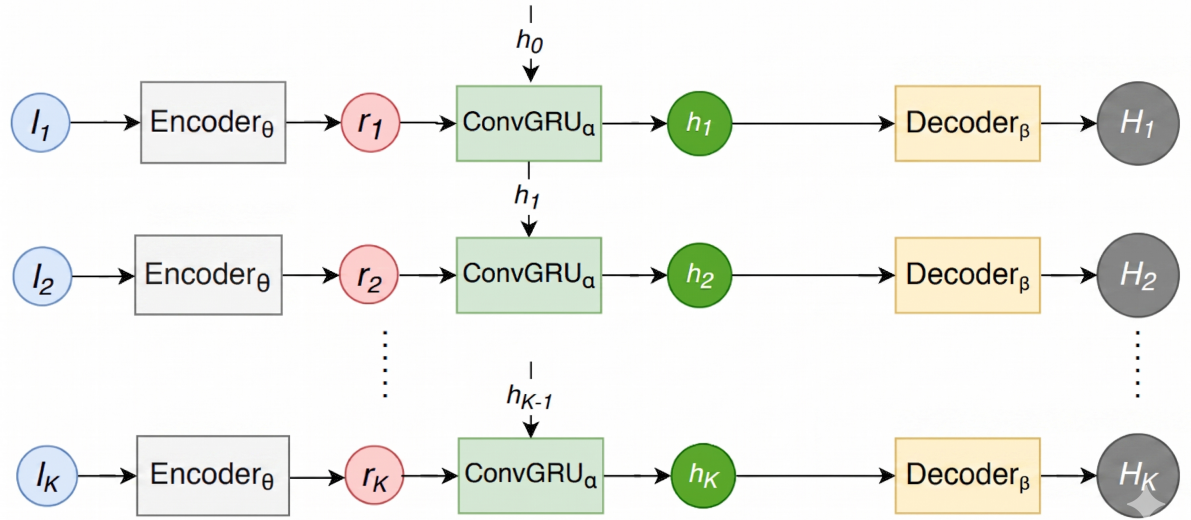


Figure 3.3: Streaming (many-to-many) MISRGRU architecture. Each input frame I_t is encoded by the shared encoder, fused by the ConvGRU with the previous hidden state h_{t-1} , and decoded to produce the SR output $\hat{Y}_t$ at every time step.

Training with Truncated Backpropagation through Time

Training on sequences of length 50 frames with standard backpropagation through time (BPTT) is memory intense: the activation graph grows linearly with sequence length. To solve this we *truncate BPTT* (TBPTT). The sequence is split into chunks of length `chunk_size`, and the hidden state is detached from the computational graph at each chunk boundary. Only gradients within a chunk are backpropagated. Moreover this helps the model to stabilize, train faster for more epochs and trains model to learn from warm state when hidden state is detached, which resembles a real-world scenario during long inferences.

A second key hyperparameter is `loss_last_n`: supervision is applied only on the *last* n frames of each chunk, allowing the GRU to accumulate meaningful context for several steps before being penalised. Supervising every frame from the first step (as in the `loss_all_frames` variant) causes the model to optimise for single-frame output quality, which prevents the GRU from learning to defer quality improvements to later frames.

Deformable Phase-space Alignment modification(DPA)

In live-cell imaging, the imaged structure can drift between frames, meaning $\mathbf{h}_{t-1}$ may encode a field of view that has shifted relative to the current encoder features $\mathbf{f}_t$. Passing a misaligned hidden state directly into the ConvGRU introduces ghosting artefacts and degrades reconstruction quality. To address this, we add a two-stage *Deformable Phase-space Alignment* (DPA) module that maps $\mathbf{h}_{t-1}$ into the coordinate system of $\mathbf{f}_t$ before the recurrent update, following the DPA-TISR[29] alignment order.

Stage 1: Frequency-domain phase correction (RFFTAlignment). Global sub-pixel drift looks as a phase shift in the 2D frequency domain. Both $\mathbf{h}_{t-1}$ and $\mathbf{f}_t$ are transformed with the real-valued FFT, and a small CNN predicts a per-channel phase correction from their concatenated phase spectra. This delta is added to the phase of $\mathbf{h}_{t-1}$ while its amplitude is kept unchanged, after which the corrected hidden state $\mathbf{h}_\phi$ is recovered via the inverse FFT.

Stage 2: Deformable convolution for local correction (DeformableConvAlignment). Residual non-rigid deformations are handled by a deformable convolution stage. A second CNN predicts per-pixel sampling offsets and from the concatenated pair $[\mathbf{h}_\phi, \mathbf{f}_t]$. The offsets warp the sampling grid of a 3×3 convolution applied to $\mathbf{h}_\phi$, while the masks gate each sample point, allowing the network to suppress unreliable aligned regions. The resulting $\mathbf{h}_{\text{aligned}}$ is passed to the ConvGRU together with the $\mathbf{f}_t$, so the SR output remains anchored to the current frame rather than a stabilised reference.

Zero-initialisation and first-frame behaviour. The output layers of both alignment CNNs and the offset/mask head are zero-initialised, and the deformable convolution weight is identity-initialised (centre tap set to one). This ensures the model starts as a standard streaming GRU with no alignment applied and learns the correction transforms progressively during training. On the first frame, when no $\mathbf{h}_{t-1}$ is available, the DPA stage is skipped entirely and the GRU is initialised from a zero hidden state. This way we can reuse trained weights and only finetune the model. Section 5.4 evaluates the reconstruction accuracy of DPA MISRGRU.

3.4.2. Motion Augmentation Training

A model trained only on static data, where every frame shows the same field of view, has no incentive to track the current position of the sample. In live-cell imaging, organelles can move at $0.5\text{--}5\ \mu\text{m/s}$ [34] depending on a structure. This corresponds to $\approx 0.05\text{--}0.5$ pixels per frame, our pixel size (100 nm) and acquisition rate of 100 fps.

To train the streaming model to be robust to motion, we introduce *motion augmentation* during training. For each training sequence, with probability $p = 0.7$ a sinusoidal drift trajectory is sampled independently for each spatial axis. Sinusoids produce smooth, bounded motion with no directional bias, and their frequency and amplitude are sampled to keep the instantaneous speed in the range $0.2\text{--}1.5$ pixels/frame. This range intentionally exceeds the biological estimate above, providing robustness to faster stage drift and to higher-fps acquisition settings where the same physical speed maps to more pixels per frame.

Augmentation is applied on-the-fly: at each training step, the 256×256 source LR frame is shifted by the trajectory displacement using bilinear interpolation, and a fixed 180×180 center crop is extracted equivalent to sliding a crop window across the source. The static HR ground truth is shifted by the same displacement scaled by the upsampling and center-cropped to the matching HR size, producing a per-frame HR target that matches the shifted field of view. The remaining 30% of training sequences use the unshifted static patches, which prevents the model from degrading on non-moving data.

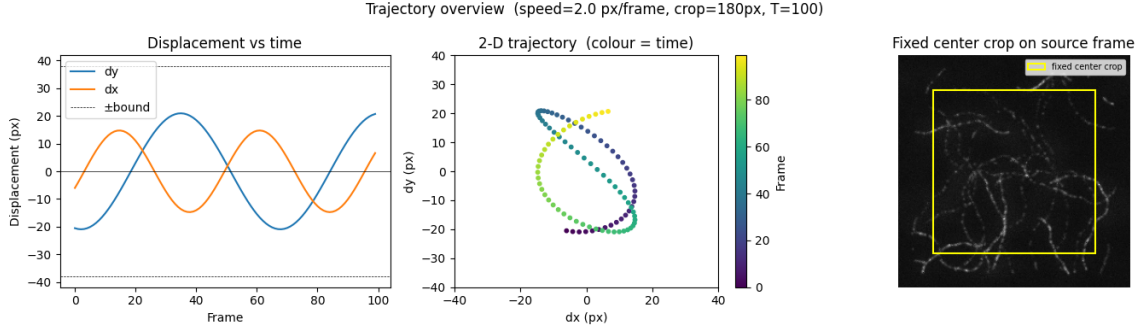


Figure 3.4: Motion augmentation. A sinusoidal drift trajectory is sampled per axis and used to shift the 256×256 source LR frame via bilinear interpolation; a fixed 180×180 center crop then slides across the source field of view. The HR ground truth is shifted by the same displacement scaled by the upsampling factor and center-cropped to the matching size, yielding a per-frame HR target aligned with the shifted view.

3.5. Training Procedure

Optimiser. All models are trained with Adam [20] ($\beta_1 = 0.9$, $\beta_2 = 0.999$). Two separate learning-rate groups are used: the encoder and decoder share a rate of $\eta_{\text{coder}} = 10^{-3}$, while the Transformer fusion module uses a lower rate of $\eta_{\text{transformer}} = 5 \times 10^{-4}$ to avoid over-updating the attention weights early in training. For the streaming MISRGRU, a single rate of $\eta = 3 \times 10^{-4}$ is used throughout.

Learning rate schedule. Three schedules were evaluated: ReduceLRonPlateau, which multiplies the learning rate by a fixed factor whenever the training loss has not improved for a set number of epochs; CosineAnnealingLR, which decays the rate from its initial value to a small minimum following a cosine curve over the full training run; and a manual step-decay, which applies the same multiplicative factor on a fixed patience counter without a torch scheduler object. A comparison run at 200 epochs showed that ReduceLRonPlateau reached a lower final loss than both cosine annealing and manual decay so strategy ReduceLRonPlateau was adopted for all subsequent experiments. The decay factor is set to 0.95 and the patience to 5 epochs.

Batch size and gradient accumulation. Each mini-batch contains 4 samples; gradients are accumulated over 4 steps before each optimiser update, giving an effective batch size of 16. Gradient accumulation is used to accommodate GPU memory constraints when training on 256×256 LR patches with 8 or 20 frames per sample in .

Training duration. TRNet models were trained for 100 epochs. For the streaming MISRGRU, we trained models on DelftBlue with limit of 24 hours. Trainings ended approximately at 190 epochs. Training logs are shown in Appendix C.

Data split. Training uses up to 2000 simulated samples from the synthetic dataset described in Chapter 4. A 90/10 random split assigns samples to training and validation sets during training. Once training is completed, a separate evaluation dataset of 12 different conditions with 40 images each is used as the test set to evaluate the performance of the models.

Hardware. TRNet experiments were run on a single NVIDIA RTX A5000 (24 GB). Training took approximately 4 and 9 hours for 8 and 20 frame models, respectively. Streaming MISRGRU experiments

were run on DelftBlue HPC cluster nodes equipped with NVIDIA A100 GPUs (80 GB), using Apptainer containers for reproducibility.

4

Dataset

This chapter describes the datasets used for training and evaluating the TRNet and MISRGRU architectures.

Origin of the simulated data. No new simulations were generated for this thesis. All experiments reuse the existing synthetic dataset produced in the RESURF paper [22]. The data were generated with the publicly available SOFI simulation tool¹ together with the SOFI package²; the simulation methodology is described in detail in [21, 22].

This work's contribution to the dataset. *Motion augmentation* of the static simulated sequences was introduced in this thesis as a preprocessing step to train and evaluate the streaming model on its ability to reconstruct moving samples. The implementation is released as part of the project codebase.³

4.1. Synthetic Data

Fluorescence microscopy experiments are time-consuming and obtaining super-resolved ground truth from real images is inherently difficult. Classical SOFI requires hundreds of raw frames to accumulate reliable cumulant statistics, meaning that acquiring a single ground-truth SR image takes tens of seconds. Over that interval, live cells drift and fluorophores irreversibly bleach, so the "ground truth" computed from the later frames reflects a different sample state than the one seen in the early frames used as input.

For this reason, a simulation-based dataset was used where conditions such as blinking speed and background noise could be controlled precisely. The synthetic dataset provides paired low-resolution (LR) fluorescence stacks of 100 frames with corresponding high-resolution (HR) SOFI targets that is used for training. Additionally, since all data was simulated we have access to the actual structure that is being imaged: Ground Truth (GT), which is then used to evaluate the structural correlation with our super-resolved prediction.

4.1.1. Simulated Data Conditions

To build a dataset that generalises across different experimental conditions, simulations were run under 12 distinct settings formed by the combination of three blinking-speed regimes and four photon-level fractions [21]:

- **Blinking speeds:** fast, medium, and slow, corresponding to fluorophore on/off-times of approximately 600ms, 1200ms, and 2000ms respectively.
- **Photon levels:** 100%, 80%, 60%, and 40% of the nominal ion photon count, progressively reducing the peak signal intensity.

¹https://github.com/GrussmayerLab/Sofi_Simulations

²<https://github.com/GrussmayerLab/sofipackage>

³Motion-augmentation code: https://github.com/GrussmayerLab/SOFI-TR-MISR/blob/motion/src/motion_augment.py.

4.2. Motion Augmentation

The synthetic dataset described above contains no sample motion: every frame in a sequence depicts the same field of view, with fluorophores at fixed coordinates emitting stochastic blinks. This is appropriate for the batch MISRGRU and TRNet models, which output a single SR image from a fixed set of N frames. The streaming MISRGRU (Section 3.4.1), however, is designed to produce a super-resolved image at *every* time step, making it directly relevant to live-cell imaging where sample motion is common and unavoidable. In practice, organelles such as vesicles and mitochondria translate across the field of view at 0.5–5 $\mu\text{m/s}$ [34]. A model trained exclusively on static data has no incentive to track these movements: since the correct answer is always the same single HR reconstruction, the network can ignore temporal position entirely. Our experiments 5.11 confirmed this where trained models produced ghosting artefacts in which fluorophore contributions from all positions were summed rather than localised to the current field of view.

Because paired moving ground-truth data cannot be obtained experimentally, all motion is synthesised from the existing static dataset. Motion augmentation converts each static training sample into a drifting sequence by shifting both the LR frames and the corresponding HR target frame-by-frame along a synthetic trajectory, forcing the streaming model to predict the HR image at the current field position. The implementation is released as part of the project codebase.⁴

An earlier approach was perturbing microtubule segment angles frame-by-frame inside the simulation software⁵ using an Ornstein–Uhlenbeck process. This would have required refactoring and re-running the full SOFI simulation pipeline for every trajectory, and was quickly abandoned for that reason; the motion augmentation described here instead operates purely on the already-simulated frames.

4.2.1. Sinusoidal Drift Trajectory

Rather than a random walk, which can accumulate arbitrarily large displacements, the trajectory is modelled as an independent sinusoid on each spatial axis:

$$\tau_a(t) = A_a \sin(2\pi f_a t + \varphi_a), \quad a \in \{y, x\}. \quad (4.1)$$

The normalised frequency is $f_a = n_a/T$, where $n_a \sim \text{Uniform}(1.5, 3.5)$ is the number of oscillation cycles over the 100-frame sequence, and $\varphi_a \sim \text{Uniform}(0, 2\pi)$ is a random phase that breaks any correlation between axes. The amplitude A_a is derived from a user-specified peak velocity v (in LR pixels per frame): since the instantaneous peak velocity of a sinusoid is $2\pi f_a A_a$, setting

$$A_a = \min\left(\frac{v}{2\pi f_a}, d_{\max}\right) \quad (4.2)$$

guarantees that the maximum drift speed equals exactly v , with $d_{\max}$ as a hard bound that prevents large zero-padded border regions. This parameterisation decouples biological realism, controlled by v , from trajectory shape, controlled by n_a and φ_a . The sinusoidal model also naturally captures the direction reversals that are characteristic of organelle motion constrained by cytoskeletal tracks.

4.2.2. Frame Shifting and Per-Frame Targets

Given a trajectory τ , each LR frame at time t is shifted by $(dy_t, dx_t) = \tau[t]$ using bilinear interpolation with zero-padding. The corresponding HR target is shifted by the same displacement scaled by the upsampling ratio, consistent with the $2n - 7$ output size.

The critical consequence is that each frame t receives its own HR ground-truth $\hat{y}_t$. In the static case, a single HR image is broadcast across all time steps, so the model can ignore temporal position entirely. With motion augmentation, the loss $\mathcal{L}(\hat{Y}_t, \hat{y}_t)$ forces the streaming model to predict the HR image at the *current* position of the field of view rather than a single static image reconstruction.

4.2.3. Relevance to Physical Speed

The peak velocity v in LR pixels per frame maps to a physical drift speed via

$$v_{\text{real}} [\mu\text{m/s}] = v [\text{px/frame}] \times r [\text{frame/s}] \times p [\mu\text{m/px}], \quad (4.3)$$

⁴Motion-augmentation code: https://github.com/GrussmayerLab/SOFI-TR-MISR/blob/motion/src/motion_augment.py.

⁵https://github.com/GrussmayerLab/Sofi_Simulations

where $r = 100$ Hz is the camera acquisition rate and $p = 0.1 \mu\text{m}/\text{px}$ (100 nm per pixel, consistent with the simulated dataset described in Section 4.1). The configured speed range $[v_{\min}, v_{\max}] = [0.2, 1.5]$ LR px/frame therefore spans $2 - 15 \mu\text{m}/\text{s}$. The biologically relevant range for vesicle and mitochondria drift is $0.5 - 5 \mu\text{m}/\text{s}$ [34], with fast kinesin-driven transport reaching up to approximately $7.5 \mu\text{m}/\text{s}$. The upper bound of $15 \mu\text{m}/\text{s}$ therefore exceeds the typical biological speeds. This way we can stress-test the model during training: by exposing the network to displacements larger than those encountered in practice, we encourage it to learn more robust motion-aware representations, with the expectation that generalisation to in-distribution speeds will be stronger as a result.

Both bounds are exposed as configuration parameters (`speed_min`, `speed_max`), making it straightforward to restrict training to slow or faster motion.

4.2.4. Integration into Training

Motion augmentation is applied on-the-fly inside the `StreamingTensorDataset` data loader. For each training sample, a motion augmentation is applied with a probability of $p_{\text{motion}} = 0.7$. This way model remembers to reconstruct static samples as well. When augmentation is applied, peak velocity is drawn uniformly from $[0.2, 1.5]$ LR px/frame, a trajectory is generated, and the frames and targets are shifted accordingly with $d_{\max} = 38$ LR px. Otherwise the static HR image is kept across all T frames.

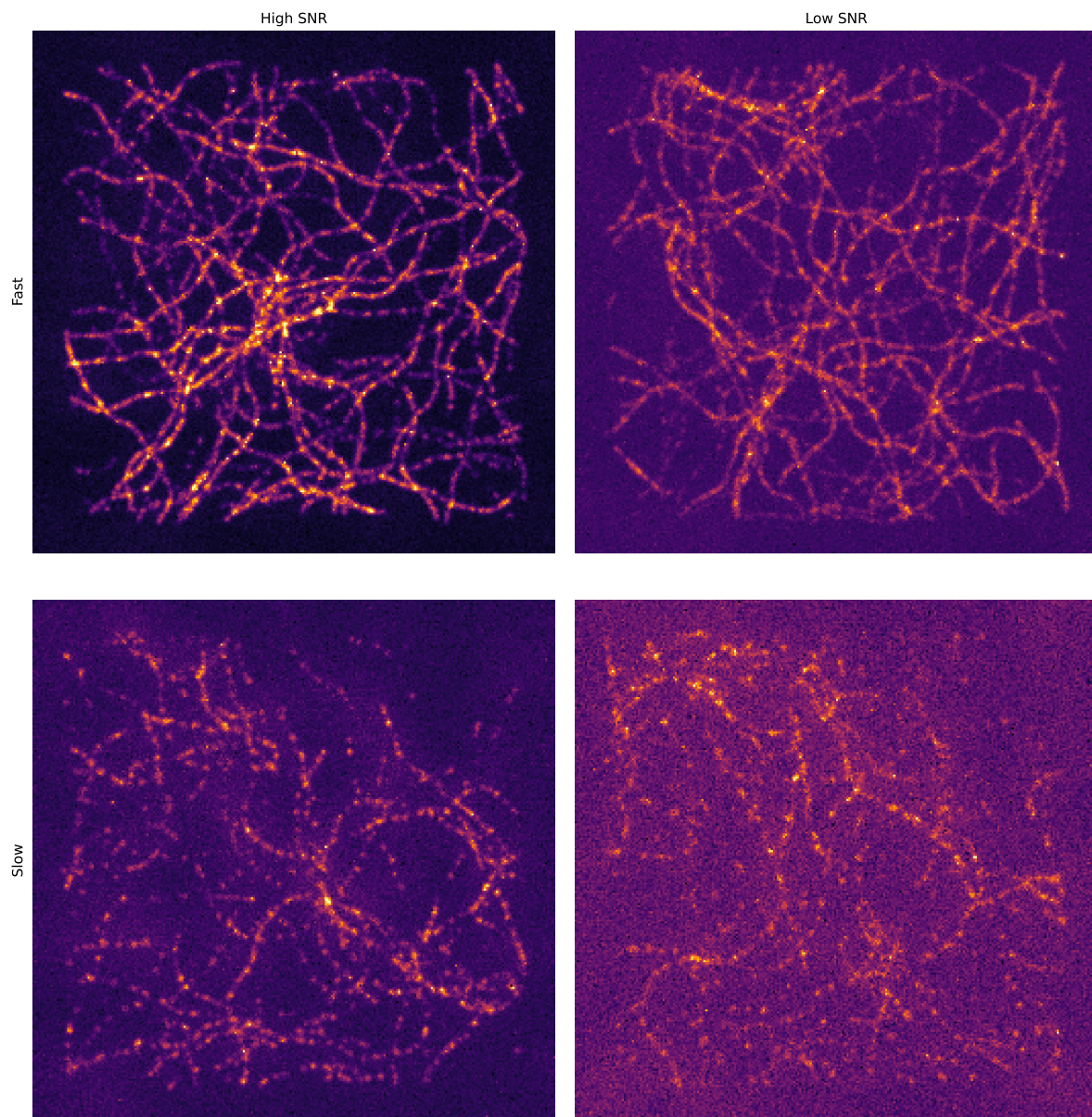


Figure 4.1: Simulated frames illustrating the different imaging conditions. Columns vary the photon level (100% to 40%), while rows correspond to fast and slow blinking speeds.

Experiments and Results

This chapter evaluates the two model families introduced earlier against the central question of whether deep-learning models can replace SOFI for fast, live-cell imaging. We first assess the TR-MISR transformer: how reconstruction quality scales with the number of fused frames, the effect of the training frame count, and the trade-off between transformer size and inference latency. We then turn to MISRGRU, modifying it for video super-resolution through a sliding-window variant and a streaming formulation that emits one super-resolved frame per input, and we study how the accumulation window governs the quality-latency balance for continuous imaging. Throughout, models are compared using metrics chosen for SOFI reconstructions (Resolution Scaled Pearson correlation and Decorrelation analysis). Encoder, decoder, loss-function and other hyperparameter choices are validated by ablation in Appendix B.

5.1. Evaluating Performance of TRMISR

5.1.1. Experimental Setup

Model Training Details

For the following TR-MISR experiments the architecture with a PixelShuffle decoder was used with a lightweight fusion module (1 attention head and depth of 1 – 1h1d), trained on 8 and 20 input frames for 100 epochs. Models are optimised with Adam using a learning rate of 10^{-3} for the encoder/decoder and 5×10^{-4} for the transformer, reduced on validation-loss plateau. The combined L1+Fourier loss is used throughout; 2000 simulated samples are used with a 90/10 train/validation split. The choice of loss function and decoder were validated by ablation; see Appendix B.

The test set comprises 12 SNR conditions spanning three blinking speeds: 10/600, 10/1200 or 10/2400 corresponding to On and Off times in milliseconds and four photon levels (100%, 80%, 60%, 40% of nominal), with 40 samples per condition (480 test samples total). For simplicity we often report only 4 conditions: fast and slow combined with highest and lowest photon levels.

Models are evaluated with the two metrics introduced in Section 2.2.6: resolution-scaled Pearson correlation (RSP) for structural fidelity and decorrelation analysis for spatial resolution. RSP is the primary metric reported in all experiments; decorrelation analysis is reported only for the best models of each class.

5.1.2. Multi-frame Fusion

Because TR-MISR’s cross-attention aggregator uses a softmax over all T input tokens, it is not architecturally constrained to a fixed frame count. A model trained on 8 frames can therefore attend to longer sequences at test time without retraining or fine-tuning. Sequential GRU fusion does not share this capability as we show in the experiment.

To verify this, we trained the TRNet with modifications as mentioned in Section 3.1. The model was trained on 8 input frames with shallow transformer size (1 head and 1 depth). During evaluation we

compared the reconstruction accuracy of using different input frames $N \in \{8, 20, 30\}$ and compare against the MISRGRU foundational models from RESURF [22].

Figure 5.1a shows RSP results using box plots calculated using 480 evaluation samples, with classical SOFI (100-frame) shown in grey on the left.

Providing more frames consistently improves TR-MISR: the 20-frame and 30-frame variants both exceed the MISRGRU 20-frame baseline in aggregate RSP shown in Figure 5.1a. Although the box plots overlap, the improvement is real rather than noise: on a matched 20-frame budget TR-MISR beats MISRGRU by 0.013 RSP on average and on roughly 80% of the paired samples (rank-biserial -0.81 , paired Wilcoxon $p < 10^{-50}$; Appendix E).

This confirms that the attention mechanism generalises to sequences longer than those seen during training.

Figure 5.2 breaks results down per SNR condition. MISRGRU is more consistent across conditions: its inter-quartile range is tighter at high-SNR settings (fast blinking, high photon counts). TR-MISR, however, degrades less at low SNR: under slow-blinking and low-photon conditions (09–12_slow), TR-MISR’s median RSP stays closer to the 8-frame MISRGRU baseline, whereas MISRGRU 8f falls more sharply.

The improvement of TR-MISR over MISRGRU is more pronounced in slow and low-SNR imaging, as shown both by the RSP metric and by visual inspection in the bottom row of Figure 5.3. We observed that MISRGRU hallucinates more often in slow blinking and low SNR conditions whereas transformer model reconstructs dim microtubules much closer to actual ground truth image.

Surprisingly, the MISRGRU 20-frame model performs worse on several slow-blinking conditions when the input is extended to 30 frames. We attribute this to the unnormalised summation in the fusion module’s GAP layer, which causes the decoder input magnitude to grow with increasing number of input frames beyond the range seen during training. Section 5.2.1 analyses this effect quantitatively.

Resolution estimation of TRNet outputs

To test how the number of input frames affects the output resolution we apply decorrelation analysis [10] to the SR outputs of TRNet trained on 8 input frames. Three numbers of input were tested: 8, 20, 30 across all four blinking regimes: fast-high SNR (FH), fast-low SNR (FL), slow-high SNR (SH), and slow-low SNR (SL).

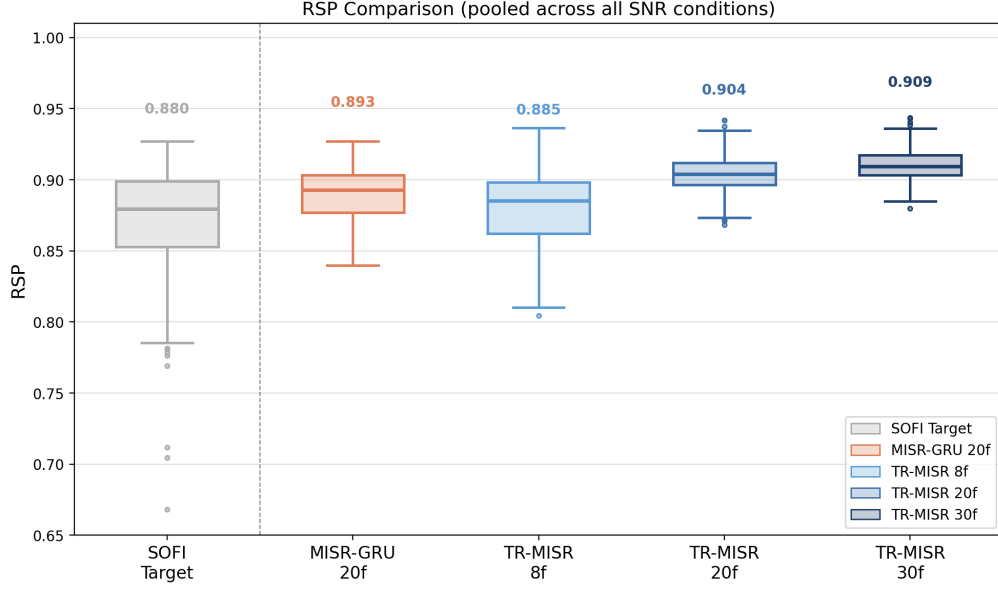
Figure 5.1b shows box plots of the estimated resolution for the model trained on 8 frames but evaluated at 8, 20 and 30 input frames. All three variations are able to resolve structure well below the optical diffraction limit, confirming genuine super-resolution across all blinking configurations, averaging $\approx 140 - 150$ nm.

The same decorrelation analysis for the RESURF [22] models and SOFI is reproduced in Figure 5.8a. There, the MISRGRU models resolve structural detail down to ≈ 130 nm, a sharpness that is also visible qualitatively in Figure 5.3: closely spaced microtubules are separated more cleanly by MISRGRU, whereas TR-MISR renders thicker filaments that score well on RSP but blur fine structure together.

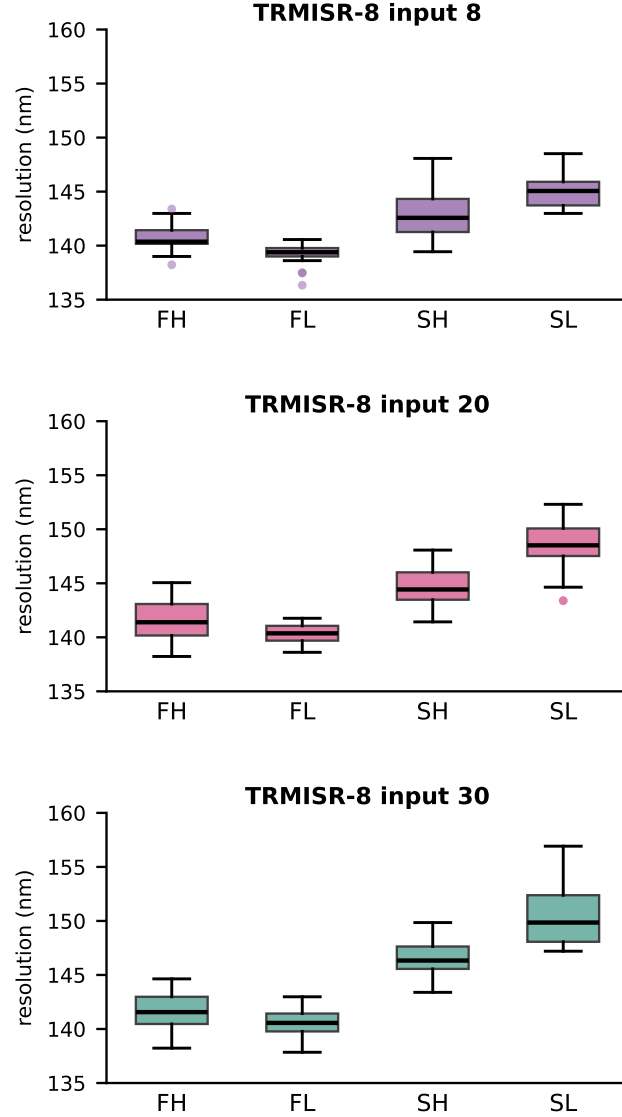
5.1.3. Effect of Training Frame Count: 8f vs 20f TRNet

To understand whether training on more frames improves TRNet, we trained an additional model on 20-frame sequences and compared it against the 8-frame trained model across $N \in \{20, 30\}$ input frames at test time. Both models were trained for 100 epochs. Loss functions are in the Appendix A Figure 5.4 shows the RSP distributions for both models.

Counterintuitively, the 8-frame trained model outperforms the 20-frame trained model when both are evaluated on 20. We believe that 8 frames is already sufficient to capture enough information produced by fluorophore blinking: at the simulated acquisition rates and blinking speeds, in 80ms time fluorophores that were on will have switched state to dark state least once within 8 frames, so the cross-attention module receives a diverse sample of blink events and can learn to aggregate efficiently. Training on 20 frames does not provide substantially more blinking statistics; it predominantly adds redundant observations of the same fluorophore states.



(a) RSP distribution for shallow TR-MISR (1 head 1 depth) trained on 8 frames and inferred on 8, 20, and 30 input frames versus the best MISRGRU 20-frame foundational model from RESURF [22]. Classical SOFI (100-frame) is shown in grey as a reference.



(b) Decorrelation analysis of TR-MISR outputs for the 8-frame trained model evaluated at 8, 20, and 30 frames across four blinking/SNR conditions. Lower resolution values indicate sharper, higher-fidelity images; all models resolve below the diffraction limit (≈ 250 nm).

Figure 5.1: Multi-frame fusion of the 8-frame trained TR-MISR evaluated at 8, 20, and 30 input frames: (a) structural fidelity measured by RSP and (b) spatial resolution estimated by decorrelation analysis, both over 480 evaluation samples.

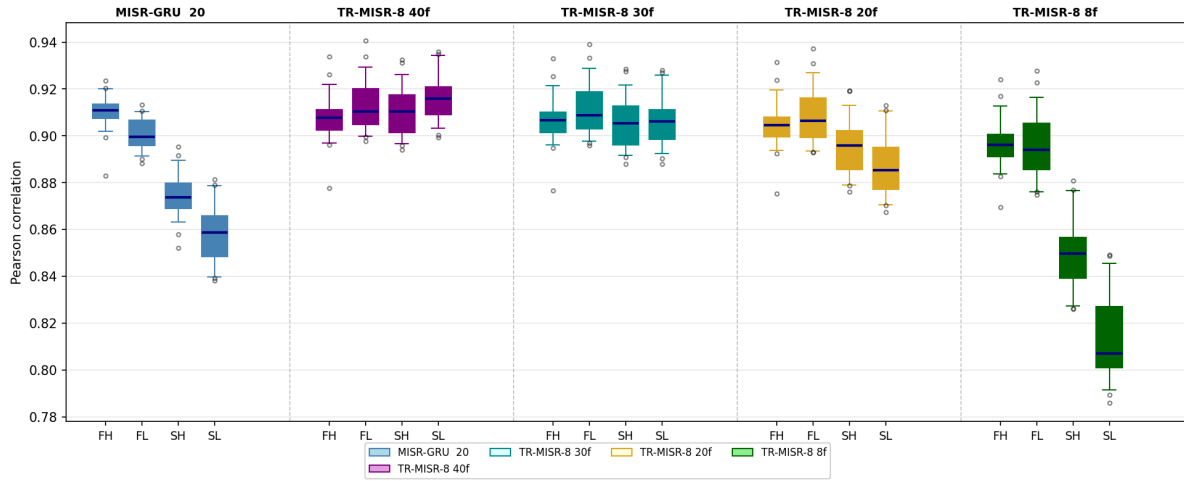


Figure 5.2: Comparison of the TR-MISR model trained on 8 frames and evaluated on 40, 30, 20, and 8 frames, alongside the foundational MISRGRU on 20 frames, broken down per SNR condition. **Take-away:** feeding more frames helps TR-MISR most in the hard slow-blinking, low-SNR conditions (right-hand groups), where it holds its quality while the MISRGRU baseline drops off.

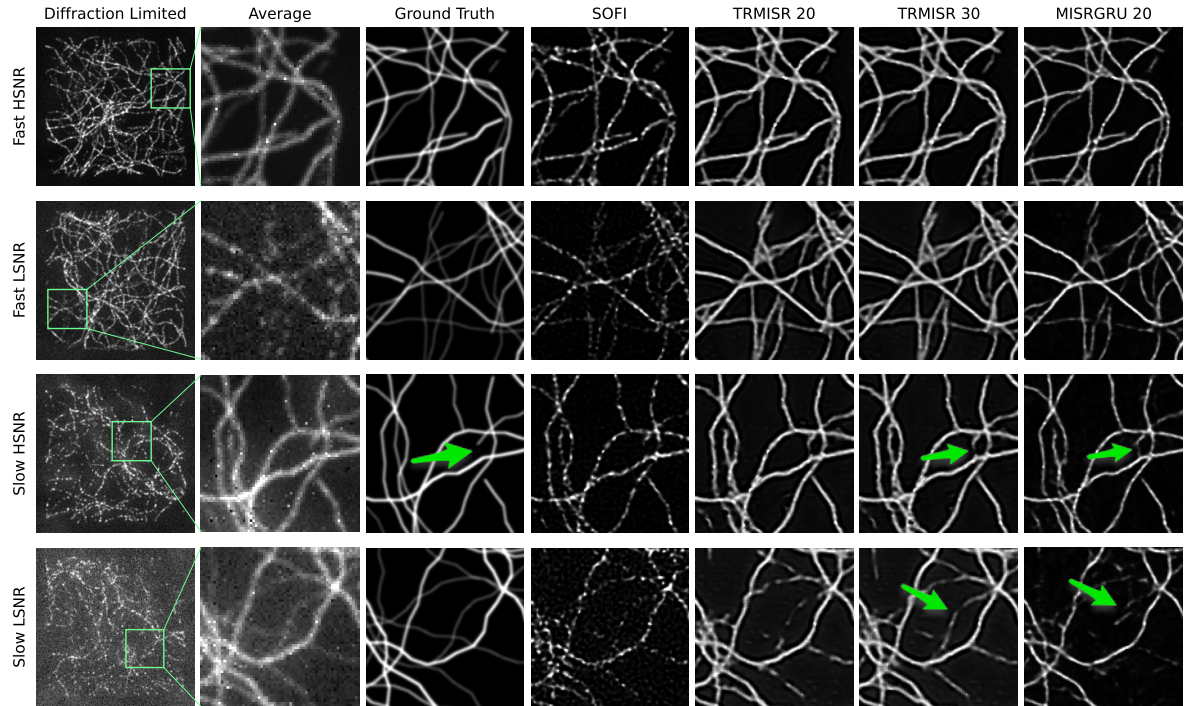


Figure 5.3: Comparison of SR reconstructions between the GT, SOFI, and TR-MISR model vs MISRGRU [22]. Single diffraction-limited frame of a simulated microtubule at different fluorophores densities and SNR levels. Each row represents different fluorophores densities and SNR levels of a microtubule structure; SR ground truth; SOFI SR image based on 100 frames; TRNet evaluated on 20 and 30 frames; MISRGRU model-based SR image based on 20 frames. Region of interest (ROI) marked by a white box, showing interesting filament reconstructions where two models differ. Bottom right MISRGRU from 20 frames hallucinates low SNR filament whereas both TRMISR models better reconstruct the microtubule

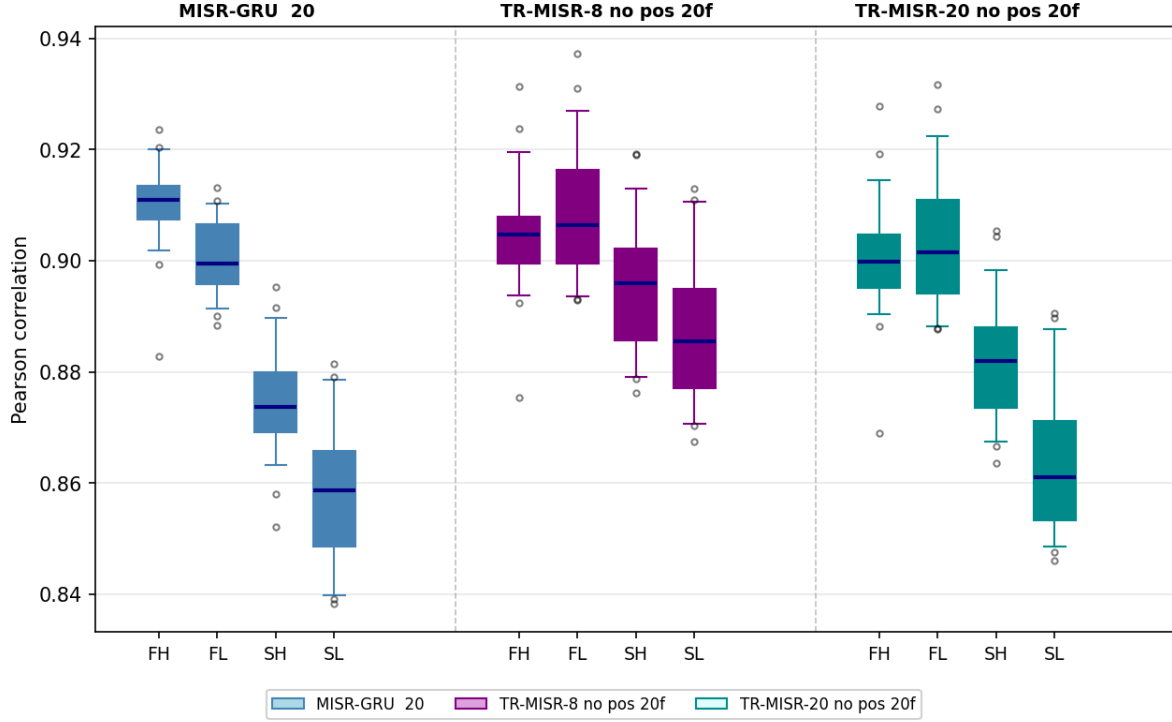


Figure 5.4: Comparison of TR-MISR model trained on 8 frames and versus TRNet trained on 20 frames, both are evaluated with 20 input frames, alongside foundational MISRGRU on 20 frames. All models are evaluated across different SNR and blinking conditions. **Take-away:** the 8-frame-trained model is at least as good as the 20-frame-trained one at every condition, so training on a short sequence costs no quality while being cheaper to train.

Pairing the two models on identical inputs, the 8-frame-trained model significantly outperforms the 20-frame-trained one on every metric (two-sided Wilcoxon signed-rank, Holm-corrected across the 12 conditions; all $p < 10^{-50}$). The matched-pairs rank-biserial correlation is ≈ 1.0 for the correlation metrics, meaning the 8-frame model wins on almost every individual sample. The advantage of the short training budget is therefore consistent across conditions, not driven by a few easy cases.

This finding reinforces the conclusion from Section 5.1.2: the attention mechanism benefits from being given more frames at inference, but the model should be *trained* on a short sequence to encourage efficient use of each frame. This way both training and finetuning can be speed up without sacrificing the reconstruction accuracy.

5.1.4. Latency vs Transformer Size Trade-offs

To figure out the best accuracy speed tradeoff of the TR-MISR model we trained four different models where we varied the sizes of heads and depths used in transformer fusion module. All four configurations converge to nearly identical validation loss. Even though the larger variants were trained for fewer epochs due to a 24 h limit on DelftBlue (Appendix D).

Figure 5.5 depicts the evaluation of different using RSP metric. Increasing the number of parameters does not improve the metric dramatically, suggesting the fusion module reaches capacity quickly and that a single attention head with one transformer layer (1h1d) is sufficient to aggregate information from multiple frames making it the preferred configuration given its substantially lower inference latency (Table 5.1).

To assess the computational cost and speed of TRNet, experiments with increasing Transformer depth and attention heads, were done. We benchmarked all four TR-MISR size variants and the MISRGRU baseline on an NVIDIA RTX A5000 (24 GB) using PyTorch and TensorRT execution. Each model was evaluated at three frame counts ($N \in \{8, 20, 30\}$) where the frame size was 512x512 pixels. 8 frames matches the MISRGRU baseline, 20 frames is the best performing foundational model in prior work,

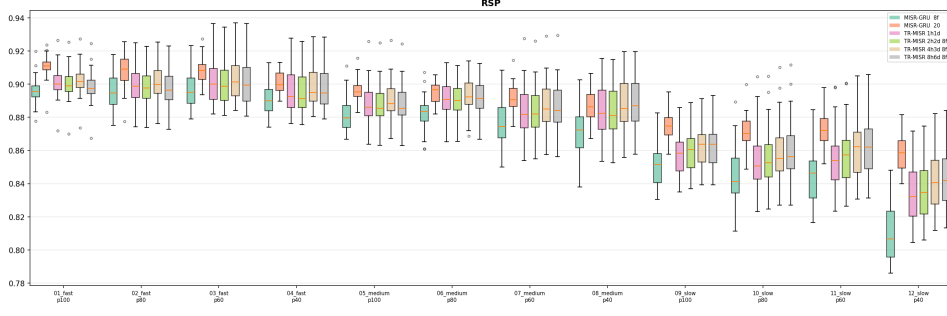


Figure 5.5: Resolution-scaled Pearson correlation for four Transformer configurations with varying depth and attention heads across SNR conditions. Baseline MISRGRUs (pre-trained 8 and 20 frames) are shown for reference. **Take-away:** RSP is essentially flat across the four transformer sizes, so the smallest (1h1d) fusion module is preferred since larger ones add latency without improving quality substantially.

and 30 frames tests scalability.

Model	Params	N=8 (ms)	N=20 (ms)	N=30 (ms)	N=40 (ms)
TR-MISR 1h1d	48,692	87.97	262.55	380.89	465.46
TR-MISR 2h2d	57,388	188.14	487.61	1200.65	-
TR-MISR 4h3d	66,084	209.10	858.60	3935.62	-
TR-MISR 8h6d	145,932	-	-	-	-
MISRGRU	95,073	67.64	170.29	255.92	-

Table 5.1: Model parameter counts and PyTorch inference latency on 512×512 images (mean over 100 runs). Measurements on NVIDIA RTX A5000 with batch size 1. Fastest inference time per frame count is highlighted in bold.

Latency scales super-linearly with transformer depth and head count: going from 1h1d to 4h3d triples the parameter count but increases N=8 latency by 2.4× (87.97 ms to 209.10 ms), while 8h6d could not be evaluated within the available memory budget. MISRGRU remains the fastest model at every frame count largely because of its purely convolutional fusion mechanism, which avoids the quadratic attention computation over the spatial sequence.

Looking at both, the RSP scores (Figure 5.5) and latency measurements (Table 5.1) lead to the same conclusion: the transformer fusion module requires minimal capacity to achieve good results. Quality does not improve substantially beyond 1h1d, however, the latency grows with model size.

5.2. Modifying MISRGRU for Video SR Reconstruction

5.2.1. Sliding-window MISRGRU Modification for Faster Inference

Subsection 5.1.2 claimed that MISRGRU oversaturates the input into the decoder and quality of the output degrades with the increasing number of frames. In this experiment we took pretrained weights from foundational models and modified the architecture to reuse the outputs h_t from convolution gru layers. We kept the weights in a queue of size 20.

The figure 5.6 shows this empirically with RSP metric as increasing number of frames are input into the model. The model is evaluated at 1, 5, 10, 20, 40, 80, and 100 input frames.

This experiment showed us that MISRGRU architecture with summative GAP layer is not well suited for multi-input-multi-output task which prompted us to change the architecture to streaming MISRGRU which normalizes the input in the decoder to produce consistent results.

The GRU hidden state h_t is a cumulative summary of all frames up to time t , so the fused representation $\mathbf{f} = \sum_{t=0}^{T-1} h_t$ is an unnormalised accumulation: the first frame contributes to every h_t , giving it T -fold weight relative to the final frame. Moreover, h_t tends to grow in magnitude, making the sum grow with T . A decoder trained on 20 input frames therefore receives activations of a different scale when the number of input frames is extended changed at test time.

To verify this claim empirically, we measured the L2-norm of the tensor passed to each model’s

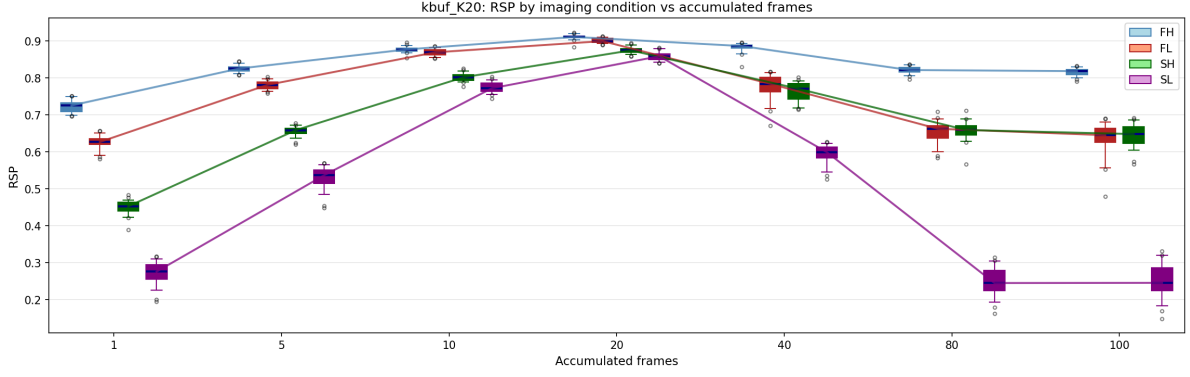


Figure 5.6: Output quality (RSP) of the sliding-window (K-buffer) MISRGRU modification as the number of input frames increases, across different SNR and blinking conditions. **Take-away:** quality peaks near the 20-frame training window and then degrades as more frames are added, exposing the unnormalised GAP summation as unsuitable for continuous, many-frame streaming.

decoder (f) on 20 real microscopy patches from the testing set, changing the number of input frames $T \in \{8, 20, 30\}$. The results are reported in Table 5.2. For both MISRGRU variants the decoder-input norm scales beyond the linear expectation, confirming the saturation hypothesis. In contrast, TR-MISR’s fusion output remains almost constant across all three frame counts (ratios of $1.02\times$ and $1.01\times$), because the cross-attention softmax redistributes weight across frames rather than accumulating magnitude. Interestingly 8 frame MISRGRU at $T = 8$ reaches approximately same L2 norm as the 20 frame MISRGRU at $T = 20$ indicating that both models decoder learnt to input similar tensor.

Table 5.2: Mean L2-norm of the decoder input f on 20 test patches (256×256) as the number of input frames T varies.

Model	Train T	$\ f\ _2 (T=8)$	$\ f\ _2 (T=20)$	$\ f\ _2 (T=30)$
MISRGRU	20	671	2 741	5 786
MISRGRU	8	2 227	8 194	12 420
TR-MISR (1h1d)	8	194	198	200

TR-MISR does not suffer from this issue because its cross-attention aggregator normalises encoded information from all T frames with a softmax. Adding more frames redistributes attention weight rather than increasing output magnitude, so the decoder always receives a input at consistent scale regardless of how many frames are input.

5.3. Streaming MISRGRU

Streaming models are evaluated by feeding frames one at a time from the test set and measuring RSP between the SR output at 20 and 100 accumulated frames compared to the Ground Truth image for 4 different imaging conditions (photons/frame): 100(high-SNR) or 40 (low-SNR), and and blinking conditions (on time/ off time in ms): 10/600 (fast) or 10/2400 (slow). The main benefit we want to test in this experiment is whether the model can achieve better reconstruction results with more accumulated images.

Experiments with ablations and hyperparameter tunings are described in appendix C. The final best performing model is evaluated in this section with following configuration:

- Loss Last N: 35 with TBPTT supervision
- Fourier-only loss
- Single 48-channel ConvGRU
- Transposed-convolution decoder

At 20 accumulated frames the streaming model already matches or slightly exceeds the batch MISRGRU baseline on every condition (FH/FL/SH/SL: Streaming 0.913/0.913/0.884/0.878 vs. Batch

0.911/0.901/0.875/0.858). Pooled over the full test set this edge is highly consistent rather than incidental: at a matched 20-frame budget the streaming model wins on almost every paired sample (mean gain 0.010 RSP, rank-biserial -0.99 , paired Wilcoxon $p < 10^{-70}$; Appendix E). With all accumulated frames from the acquisition (100 frames) RSP rises to 0.929/0.927/0.916/0.888, a consistent gain over the 20-frame batch model that is more pronounced in the harder slow and low-SNR conditions (Figure 5.7). The batch MISRGRU cannot benefit in this way, as its fixed 20-frame input window and GAP fusion mechanism prevents it from integrating additional context and TRMISR is limited by its compute and memory. The largest RSP gain, from 0.875 to 0.916, occurred under slow blinking and high SNR. This is expected: under slow blinking each emitter’s on/off cycle is long relative to the accumulation window, so 20 frames undersample the fluctuation statistics, and additional frames capture more independent blinking events and yield a better cumulant estimate. (Figure 5.9).

Decorrelation analysis estimates the ground-truth images themselves at approximately 139 nm, which sets the resolution ceiling that the model could reach if it reconstructed the image perfectly. Accumulating more frames improves both metrics, but to differing degrees: RSP improves substantially (Figure 5.7), while the corresponding resolution estimates (Figure 5.8) improve only slightly relative to outputs reconstructed from 20 frames.

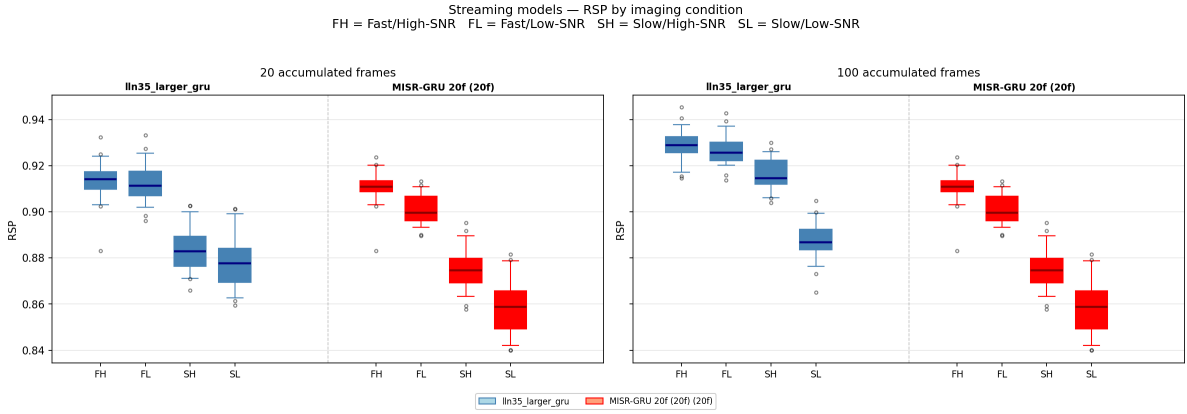


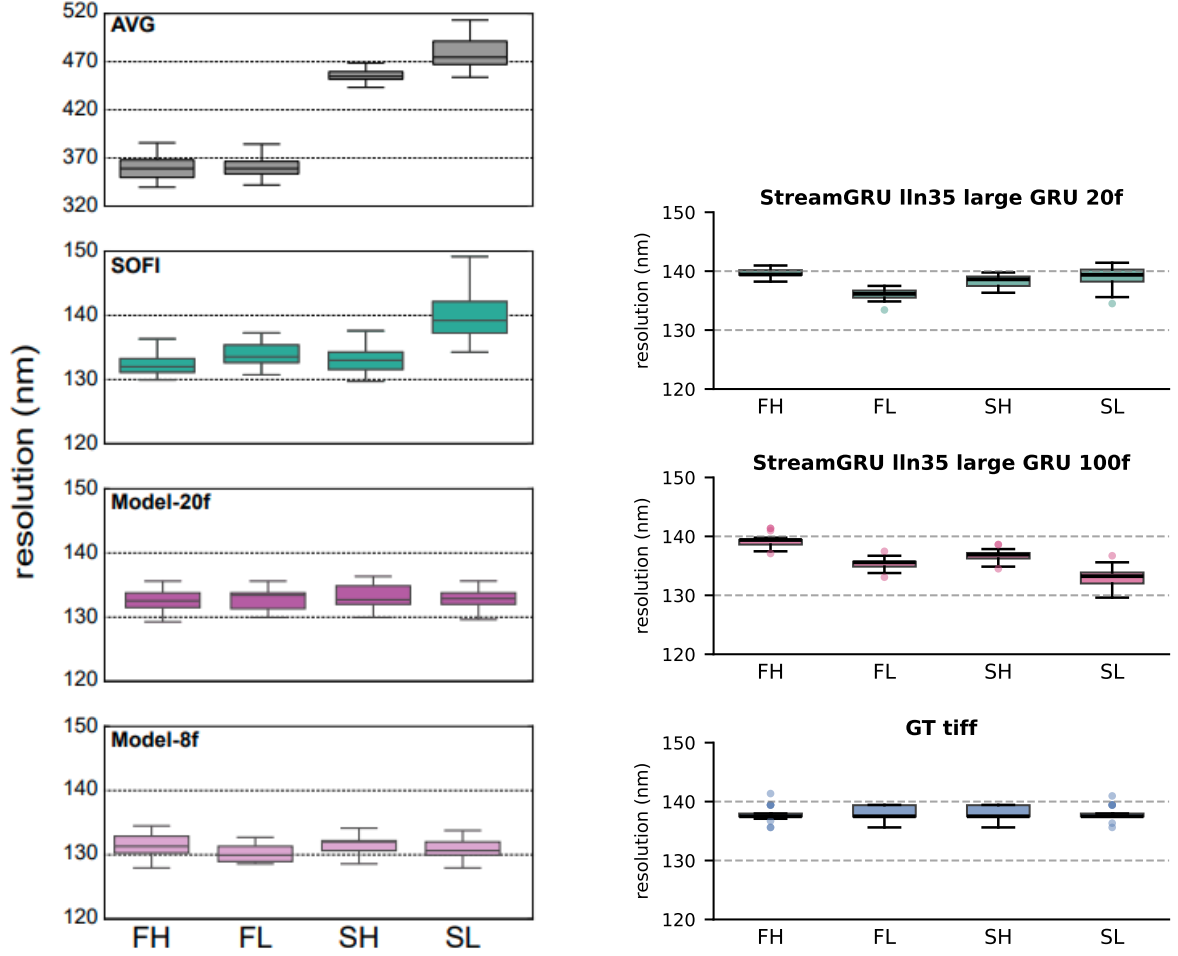
Figure 5.7: RSP performance of streaming MISRGRU model compared to batch MISRGRU (20 frames). Each column corresponds to a different SNR (photons/frame): 100 (high-SNR) or 40 (low-SNR), and blinking conditions (on time/ off time in ms): 10/600 (fast) or 10/2400 (slow). Corresponded resolution estimations and RSP scores of the models for different conditions. RSP analysis is compared with reference ground-truth (GT), a FH: Fast high SNR, FL: Fast low SNR, SH: Slow high SNR, SL: Slow low SNR. **Take-away:** the streaming model matches the batch baseline already at 20 frames and keeps improving with more accumulated frames, with the largest gains in the harder slow-blinking, low-SNR conditions.

5.3.1. Finding the Optimal Accumulation Window

Accumulating more frames improves reconstruction quality, but at the cost of temporal resolution, since each additional frame delays the moment a fully-informed reconstruction becomes available. To find the point at which this trade-off saturates, we fed the streaming MISRGRU sequences of up to 500 frames and recorded the RSP of the output after $N \in \{8, 20, 50, 75, 100, 200, 300, 400, 500\}$ accumulated frames. Because each simulated acquisition contains only 100 frames, the sequence is looped once all 100 frames have been ingested. The experiment was run on the four corner SNR conditions (FH, FL, SH, SL); Figure 5.10 reports the resulting RSP distributions.

Across all conditions, quality improves with first ~ 50 frames and then plateaus. Beyond this peak, quality no longer improves and the slow conditions even decline slightly: SL falls from ~ 0.90 at 75 frames to ~ 0.85 by 200 frames and stays there, while fast blinking dataset settles just below its peak. The fast conditions hold their plateau out to 500 frames. This decline in the high-frame regime should be read with caution, since past 100 frames the looped sequence re-presents already-seen frames rather than new fluctuation statistics.

Taken together, accumulating roughly 50–75 frames captures essentially all of the attainable quality while keeping the temporal resolution as high as possible, making it the preferred operating point for the streaming model.



(a) Resolution estimations of foundational MISRGRU models, averaged wide-field, and SOFI for different conditions. Figure taken from RESURF[22] paper.

(b) Resolution estimations of Ground Truth images and our Streaming MISRGRU trained model (evaluated at 20 and 100 frame outputs) across SNR conditions.

Figure 5.8: Side-by-side decorrelation analysis: RESURF[22] baseline models (left) vs. our models (right). Each row corresponds to a different SNR (photons/frame): 100(high-SNR) or 40 (low-SNR), and blinking conditions (on time/ off time in ms): 10/600 (fast) or 10/2400 (slow). Lower resolution values indicate sharper super-resolved images. **Take-away:** unlike RSP, the decorrelation-estimated resolution improves only marginally when more frames are accumulated, indicating that added frames mostly refine structural fidelity rather than the achievable resolution.

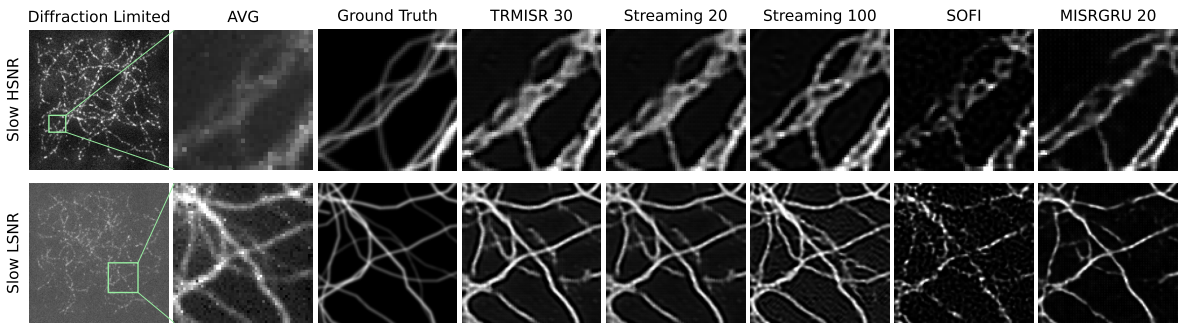


Figure 5.9: Qualitative comparison of the streaming MISRGRU, the batch MISRGRU, TR-MISR, and 100-frame SOFI against the ground truth across the four corner SNR conditions. **Take-away:** accumulating more frames lets the streaming model recover dim, slow-blinking filaments that the 20-frame batch model leaves blurred or missing, matching the quantitative RSP gains in Figure 5.7.

Notably, this saturation point coincides with the training window: the streaming model was supervised only on the first 50 frames of each simulated acquisition, so it is hard to tell whether the model is genuinely unable to integrate information beyond 50 accumulated frames or whether the cumulant estimate itself has simply saturated.

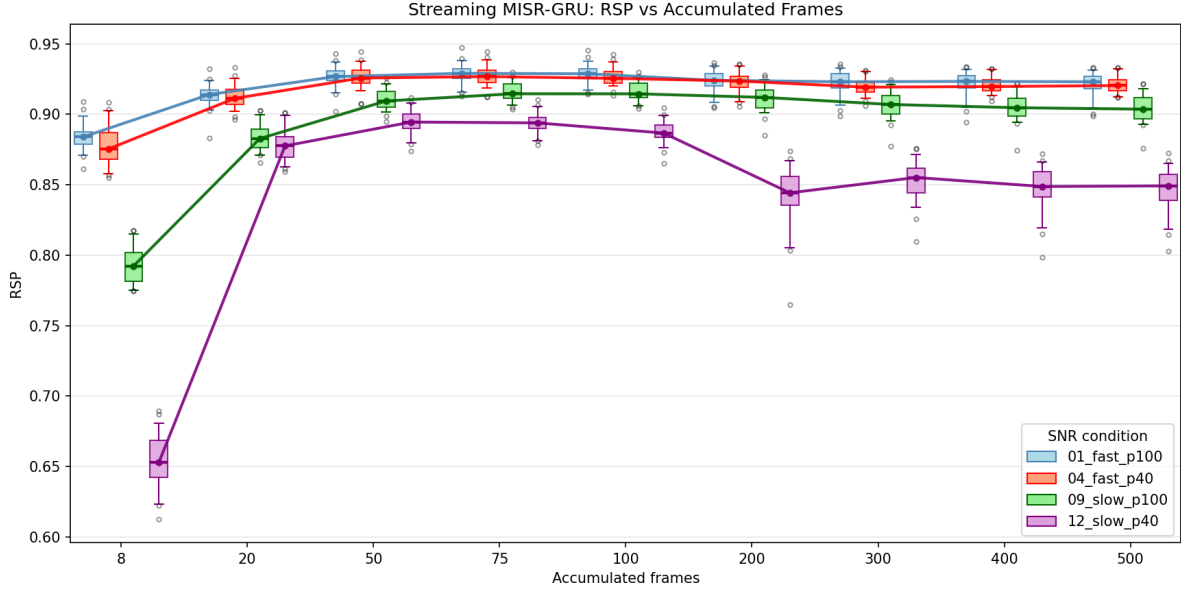


Figure 5.10: RSP of the streaming MISRGRU as a function of the number of accumulated input frames, for the four corner SNR conditions (FH: 01_fast_p100, FL: 04_fast_p40, SH: 09_slow_p100, SL: 12_slow_p40). Quality rises steeply over the first ~50 frames and then saturates; the slow conditions show a mild decline once past their peak.

5.4. Motion reconstruction

The low per-frame latency of the streaming MISRGRU makes it a natural candidate for imaging samples that move during acquisition, where the static-scene assumption underlying SOFI breaks down. We therefore evaluated how well the streaming models reconstruct moving microtubule structures.

To test whether explicit inter-frame alignment helps, we trained a streaming variant incorporating the Deformable Phase-space Alignment module (DPA) inspired by [29]; implementation details are given in subsection 3.4.1. Both motion variants were initialised from the converged fourier baseline streaming MISRGRU weights (Figure C.3) and finetuned on motion-augmented data. The plain motion-finetuned model (Finetune-motion) was trained for 100 epochs; because the DPA variant converged faster and reached higher reconstruction quality, we trained it with lln35 weights (Figure C.1) for 169 epochs (DPA). The remaining three rows of Table 5.3 are unmodified, static-trained baselines included as references: the larger streaming GRU (LLN35-larger-GRU), the batch MISR-GRU, and the batch TR-MISR.

We evaluated reconstruction accuracy under three acquisition speeds (static, $\approx 2.0 \mu\text{m/s}$, and $\approx 5.0 \mu\text{m/s}$) across the four corner SNR conditions (FH, FL, SH, SL).

Table 5.3 reports the RSP at 20 accumulated frames. The motion-augmented DPA streaming variant retains strong reconstruction quality on both static and moving samples, whereas the static-trained baselines collapse. At $2.0 \mu\text{m/s}$ and fast blinking the DPA model on moving data matches its own static performance and even exceeds the static TR-MISR baseline (93.3 ± 0.7 vs. 90.7 ± 0.8), while the static-trained TR-MISR and MISR-GRU fall below 73 at the same speed. As expected, absolute quality is lower in the slow-blinking conditions (SH, SL) for every model, since fewer fluorophores are simultaneously ON and each accumulated frame carries less information about the underlying microtubule structure.

The visual comparison in Figure 5.11 mirrors these numbers. The static-trained TR-MISR and batch MISR-GRU cannot reconstruct the moving sample: they accumulate previously seen blinks without resolving the underlying structure, so the output resembles an averaged blur. Both motion-finetuned streaming models recover fine detail the DPA variant additionally shows slightly less ghosting, visible in

the top part of the image.

Table 5.3: RSP $\times 100$ at 20 accumulated frames across three motion conditions. FH = fast/high-SNR, FL = fast/low-SNR, SH = slow/high-SNR, SL = slow/low-SNR. Best value per condition in bold.

Model	FH	FL	SH	SL
<i>Static (0.0 px/frame)</i>				
DPA	92.7 $\pm$ 0.7	92.3 $\pm$ 0.7	88.4 $\pm$ 0.9	84.6 $\pm$ 1.2
Finetune-motion	88.9 $\pm$ 1.0	88.4 $\pm$ 1.1	84.3 $\pm$ 1.5	76.9 $\pm$ 1.6
LLN35-larger-GRU	91.5 $\pm$ 0.6	91.1 $\pm$ 0.7	88.4 $\pm$ 1.0	87.7 $\pm$ 1.1
MISR-GRU	91.2 $\pm$ 0.5	90.0 $\pm$ 0.6	87.6 $\pm$ 0.9	85.8 $\pm$ 1.0
TR-MISR	90.7 $\pm$ 0.8	90.5 $\pm$ 0.9	89.6 $\pm$ 1.2	88.5 $\pm$ 1.2
<i>Moving (0.2 px/frame $\approx$ 2.0 μm/s)</i>				
DPA	93.3 $\pm$ 0.7	92.7 $\pm$ 0.9	89.5 $\pm$ 1.2	85.7 $\pm$ 1.7
Finetune-motion	86.3 $\pm$ 1.5	83.5 $\pm$ 2.2	73.4 $\pm$ 2.5	65.4 $\pm$ 2.5
LLN35-larger-GRU	69.2 $\pm$ 6.5	67.5 $\pm$ 7.1	62.8 $\pm$ 6.9	57.9 $\pm$ 7.3
MISR-GRU	69.0 $\pm$ 6.8	64.5 $\pm$ 7.7	60.5 $\pm$ 7.4	50.5 $\pm$ 7.6
TR-MISR	72.3 $\pm$ 6.1	70.4 $\pm$ 6.5	68.9 $\pm$ 6.5	63.4 $\pm$ 6.6
<i>Moving (0.5 px/frame $\approx$ 5.0 μm/s)</i>				
DPA	92.9 $\pm$ 0.9	92.3 $\pm$ 1.0	89.1 $\pm$ 1.4	84.8 $\pm$ 1.6
Finetune-motion	85.4 $\pm$ 1.5	81.9 $\pm$ 2.0	69.9 $\pm$ 2.4	60.5 $\pm$ 2.2
LLN35-larger-GRU	55.9 $\pm$ 7.0	53.5 $\pm$ 6.9	46.3 $\pm$ 7.2	38.2 $\pm$ 6.8
MISR-GRU	54.6 $\pm$ 8.2	49.1 $\pm$ 8.2	43.0 $\pm$ 8.3	29.6 $\pm$ 6.5
TR-MISR	63.2 $\pm$ 6.4	62.2 $\pm$ 6.2	58.0 $\pm$ 7.2	49.8 $\pm$ 6.6

5.4.1. Streaming Models Inference Latency

The architectural distinction between batch and streaming inference has a direct consequence on throughput: the batch MISRGRU consumes N frames and outputs a single SR image, whereas the streaming variant emits one SR output for every input frame. Models were benchmarked on an NVIDIA RTX A5000 using 512×512 input frames, 5 warmup passes, and 20 timed passes.

Model	Params	Total / seq (ms)	SR outputs	Per SR frame (ms)
Streaming MISRGRU	95,073	225.90 $\pm$ 0.42	20	11.29
Streaming Larger GRU	131,361	240.44 $\pm$ 0.67	20	12.02
Streaming MISRGRU DPA	151,996	349.26 $\pm$ 0.86	20	17.46
Batch MISRGRU (ref.)	95,073	170.29	1	170.29

Table 5.4: Streaming model parameter counts and PyTorch inference latency on 512×512 images (mean over 20 runs, sequence length $N = 20$). The *per-frame* column divides the total sequence time by the number of SR outputs produced. Batch MISRGRU (reproduced from Table 5.1) is included as a reference; it produces only one SR output from 20 input frames.

All three streaming variants produce reconstructed outputs far more efficiently per frame: they match the accuracy of the batch MISRGRU by the 20th frame and continue to improve as more frames accumulate, while emitting one SR image per input frame rather than a single output after the full sequence. The base streaming MISRGRU produces each SR frame in only 11.29 ms and the DPA variant in 17.46 ms. Processing the full 20-frame sequence is nonetheless slower than the batch model (226 ms and 349 ms versus 170 ms), because the decoder is invoked once per frame rather than a single time at the end.

Both variants stay well under the 25 ms budget required for real-time reconstruction at 40 fps, making the streaming MISRGRU an efficient choice for reconstructing an entire acquisition continuously, frame by frame, without the need to restart the model. Since the DPA aligns the frames together we achieve temporal-resolution equal to camera acquisition rate.

Decorrelation analysis results closely match the results of statically trained streaming MISRGRU model. Both models achieved better resolution at 100 accumulated frames (Figure 5.12).

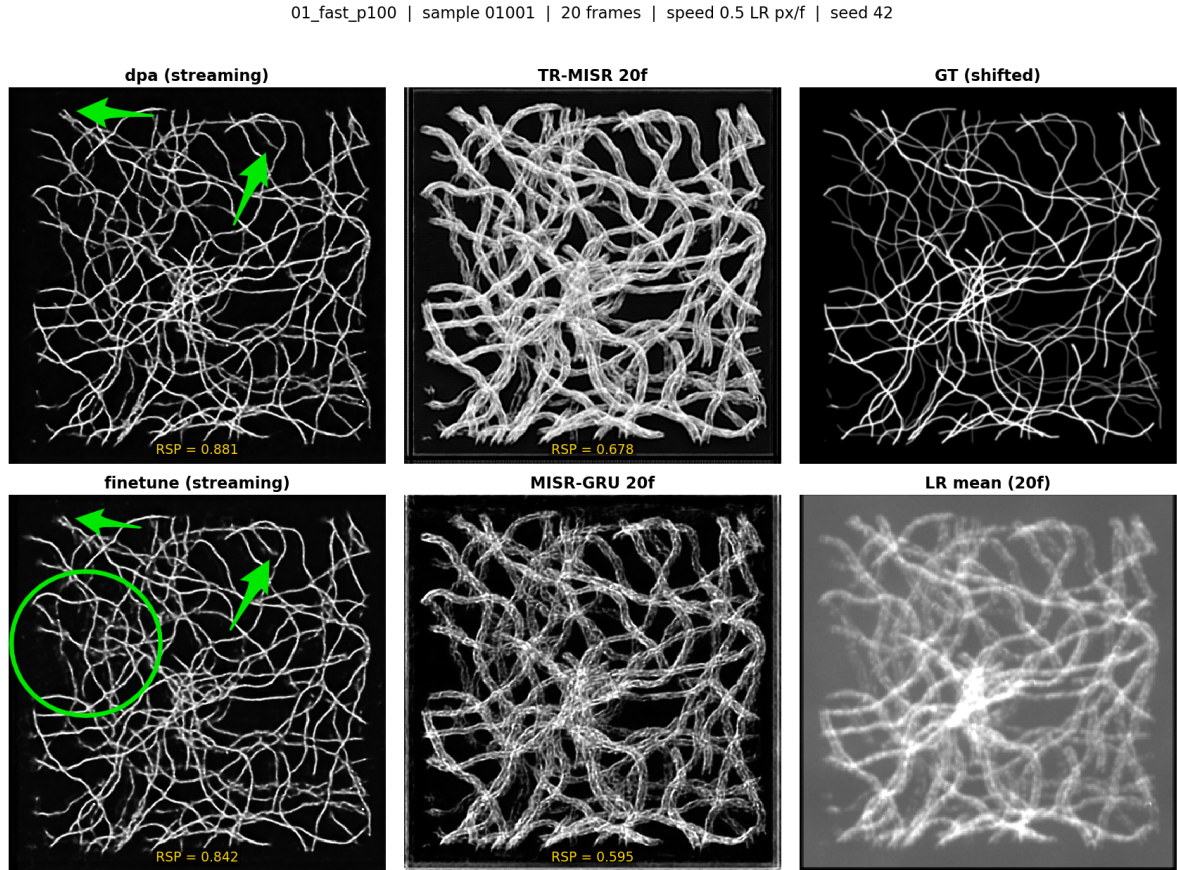


Figure 5.11: Visual comparison of finetuning streaming MISRGRU and DPA variant in motion reconstruction. The figure demonstrates the preservation of microtubule structures under $5 \mu\text{m/s}$ speed augmentation. **Take-away:** both motion-finetuned models recover the moving structure, but the DPA variant leaves fewer residual blur and ghosting artefacts; the green arrow marks a blurred region in the plain finetuned output that DPA reconstructs more cleanly.

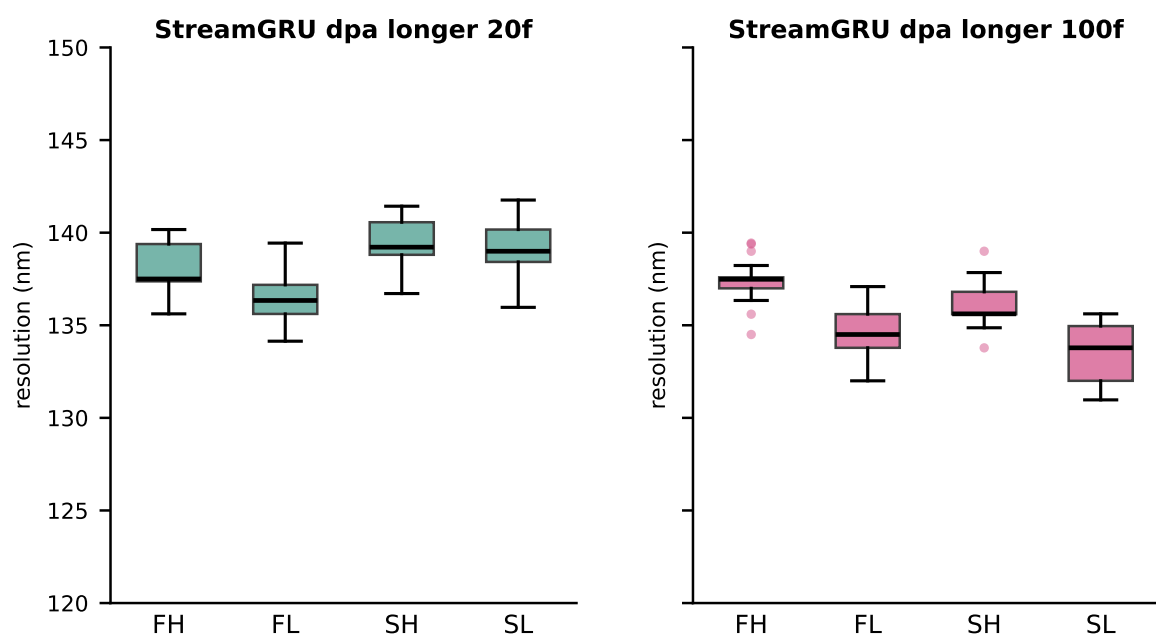


Figure 5.12: Decorrelation analysis of static DPA model at 20 and 100 frames reconstruction. Each row corresponds to a different SNR (photons/frame): 100(high-SNR) or 40 (low-SNR), and blinking conditions (on time/ off time in ms): 10/600 (fast) or 10/2400 (slow). Lower resolution values indicate sharper super-resolved images. **Take-away:** the motion-trained DPA model reaches the same resolution as the static streaming model, so adding motion robustness into training does not cost spatial resolution on static samples.

6

Discussion

Having real-time feedback of the reconstruction matters a lot for live-cell imaging because it allows biologists to observe the live sample as it interacts with the world and intervene if something goes wrong. Ideally, a smart microscope could act on this feedback on its own, adjusting the acquisition without an operator in the loop. In a post-processing workflow the low-resolution frames are collected and reconstructed only afterwards, so failures such as focus drift, photobleaching, an incorrect exposure, or the cell moving out of the field of view may go unnoticed until the session is over and the data is analysed. A streaming reconstruction overcomes these problems while the experiment is still running, allowing the operator to refocus, reposition the sample, adjust the imaging parameters before the data quality degrades, or abort an unproductive acquisition early. This whole thesis aimed towards how could such streaming reconstruction be implemented.

6.1. Does parallel attention improve upon recurrent fusion?

The main motivation of this work was to test whether replacing the sequential ConvGRU fusion in MISRGRU [22] with a Transformer’s parallel self-attention would yield a meaningful improvement in either reconstruction quality or inference speed: In aggregate resolution-scaled Pearson correlation, TRMISR with a shallow transformer (1h1d) is competitive with but does not consistently outperform the 20-frame MISRGRU foundational model on fast blinking and high SNR conditions. However the attention advantage starts to show when models are evaluated on worse conditions where TRMISR achieved consistently higher RSP. This finding is consistent with our hypothesis that transformer fusion is more robust to noisy conditions. But on raw latency it is slower (262 ms vs. 170 ms at $N=20$, Table 5.1). Therefore, the attention mechanism does not give a clear answer whether to use it over convolutional recurrence as it depends on acquisition conditions such as blinking and signal-to-noise ratio and compute constraints.

Practical advantage of TRMISR is that the model trained on 8 frames can be inferred on 20, 30 or 40 frames (if the memory allows it) at test time with no architectural change, whereas extending MISRGRU beyond its training length causes progressive reconstruction degradation. TRMISR’s the cross-attention softmax normalises over all input tokens, so the magnitude of the fused representation remains stable regardless of how many frames are presented at inference (Table 5.2). Batch MISRGRU’s summative GAP layer accumulates and oversaturates which degrades the output of the network removing the ability of feeding more input frames at inference time.

This property is valuable in practice as it means that the model can be trained on shorter training sequences reduce compute cost and convergence time, while the researcher still has the option of feeding more frames for improved quality at inference.

An additional finding reinforced this: Counterintuitively, a TRNet trained on 8 frames outperforms one trained on 20 frames when both are evaluated on 20- or 30-frame inputs (Figure 5.4). We believe this happened because the training under the 8-frame constraint forces the attention module to assign meaningful and more discriminative weights to every available frame.

Ablating the Transformer fusion across four configurations (1h1d, 2h2d, 4h3d, 8h6d) shows that RSP does not improve meaningfully beyond 1h1d (Figure 5.5), while latency grows steeply: going from 1h1d to 4h3d triples the parameter count but increases $N=8$ latency by 2.4 $\times$. Both enlarging the Transformer fusion module and widening the encoder do improve reconstruction quality, but only at a steep latency cost: the gains are statistically significant yet small in mean RSP, while latency grows several-fold.

A natural architectural extension is a streaming TRMISR. This thesis pairs a Streaming MISRGRU with a batch Transformer, but never a streaming Transformer, leaving one cell of the design space unexplored. Since attention degraded more gracefully than recurrence under the harder blinking and low-SNR conditions, so carrying that robustness into the continuous-output regime is promising. An attention variant with a key-value cache that emits one output per frame could combine the reconstruction robustness of TRMISR with the throughput of the streaming GRU. Finally, the motion augmentation developed here should transfer: pairing such a model with motion-augmented training and a DPA module could extend its robustness from blinking and low-SNR conditions to moving samples as well.

6.2. Towards Video Super-Resolution - Streaming MISRGRU

With TRMISR's substantially slower inference speed, we saw clearer path to try optimize the MISRGRU for real-time video super-resolution during microscope acquisitions. We explored reusing weights from foundational MISRGRU with sliding-window approach, quickly realizing the GAP bottleneck as well as the hidden layer should already encode most of the features.

The batch model produces one SR output per N acquired frames; the streaming model produces one output per frame, amortising the processing cost over the entire N Frame acquisition. At $N=20$ frames on a 512×512 field of view, streaming MISRGRU takes 11.29 ms using PyTorch per SR output compared to 170.29 ms for the batch model a 15 $\times$ improvement in per-output latency. The latency results reported here are obtained with PyTorch inference. Compiling the models with TensorRT and applying FP16 or INT8 quantisation could potentially half the inference times.

Streaming MISRGRU RSP converges to the batch-model value by the 20th accumulated frame (Figure 5.10), and continues to improve beyond that point as the GRU accumulates more context.

We found that the reconstruction quality saturates at around 50 accumulated frames, which sets the practical operating point for the streaming model. At a 100 fps acquisition rate this corresponds to a temporal resolution of 0.5 s. Because the model emits one super-resolved output for every frame it receives, the reconstruction also captures motion implicitly: a structure that moves over the accumulation window appears blurred in the output, signalling the presence and direction of motion to the operator.

6.3. Motion reconstruction robustness

Real-time imaging of moving structures was achieved with motion-augmented training. We compared finetuning the baseline model against adding a Deformable Phase-space Alignment (DPA) module. The final DPA model proved effectively motion-invariant: at fast/high-SNR its RSP held near 0.93 from static through $5 \mu\text{m/s}$, while the static-trained streaming, batch MISRGRU and TR-MISR baselines collapsed (below 0.73 at $2 \mu\text{m/s}$ and 0.55 at $5 \mu\text{m/s}$) because they fuse spatially misaligned frames into a blur. Plain finetuning also handled motion but trailed DPA by several RSP points and, on visual inspection, left more blur and ghosting artefacts while converging more slowly.

Importantly, the motion-finetuned DPA model outperforms the statically trained Streaming MISRGRU on both static and motion augmented evaluation dataset. TRMISR still achieves higher reconstruction accuracy on slow blinking and low signal-to-noise ratio images; this crossover between DPA and TR-MISR is significant when compared for each blinking and SNR condition separately (Appendix E).

Degradation concentrated in the slow conditions where temporal averaging is most valuable. This trade-off is expected: a model trained to handle inter-frame displacement cannot simultaneously exploit the long-range temporal correlation that static, slow-blinking acquisitions provide. For deployment, the static streaming model is preferred when the sample is known to be stationary or the drift is below the model's tolerance; the motion-finetuned or DPA variant should be selected when the sample is expected to exhibit visible drift relative to the field of view.

6.4. Reflection in the broader DSAIT context

This section places the work in the wider Data Science and Artificial Intelligence Technology (DSAIT) field and discusses what it means for the people who would use it.

6.4.1. Relation to the broader field

At a higher level, this thesis fuses a temporal sequence of observations into a single high-quality output under a tight latency budget, a problem that appears well beyond fluorescence microscopy. The central question, whether parallel attention should replace recurrent processing, repeats the shift the field went through when Transformers replaced recurrent networks in language and vision. Here that shift did not carry over cleanly as recurrent architecture achieved lower latency and better results on high signal-to-noise data.

The work also reuses an architecture built for one domain in a very different one, a common pattern in DSAIT. This required real adaptation and domain knowledge rather than drop-in reuse, which shows that such transfer is rarely free. Finally, the work treats inference latency and deployability as primary goals, placing it in the area of efficient machine learning, where the aim is comparable quality in real time on ordinary hardware rather than a higher accuracy number. This matters as AI moves from offline benchmarks into live systems.

6.4.2. Implications, applicability, and stakeholders

The direct users, the biologists operating the instrument, benefit most: real-time feedback turns a passive recording into an interactive process and lets them correct or abandon a measurement before it is wasted. The same properties open up applications beyond passive observation. In image-based cell sorting, for instance, cells must be classified on the fly from a handful of frames before a sorting decision is made; a model that super-resolves from only a few frames and corrects for sample drift could supply sharper, motion-stabilised images within the sorting latency budget, improving the discrimination of subcellular phenotypes that a diffraction-limited frame cannot resolve.

The main risk of a generative model is hallucination: it can produce plausible structure that the measurement does not support, and it does so most in the difficult conditions where the output is hardest to check. If such an artefact is mistaken for real signal, it can corrupt a scientific conclusion or a clinical decision. Responsible use therefore relies on the safeguards discussed in Section 6.5: report uncertainty with the output, be clear about the limits of the validation, and treat anything visible only in the model output as something to confirm independently. The tool should be presented as a fast approximation of a trusted method, not as a replacement for verification.

6.5. Limitations and Future Directions

First, all quantitative results are obtained on a synthetic microtubule dataset generated with the simulation pipeline from [22]. Transfer to real experimental data is not addressed in this thesis. The most immediate direction for future work is to establish whether the models can reconstruct live-cell acquisitions.

A one promising direction for adapting synthetically trained models to a specific microscope is to fine-tune the network on a sequence of blank background frames, that capture only detector noise and autofluorescence. This way the model could quickly adapt to noise levels [24], and extract the signal easier.

Once the models prove capable of reconstructing live-cell data, whether through fine-tuning or out of the box, a natural next step is to pursue general-purpose super-resolution. Most current methods [29, 22] must be fine-tuned for the specific sample being imaged, or different PSF. We believe that deep-learning models could instead operate like SOFI, generalising across acquisitions without being retrained for a particular point-spread function or sample structure.

A further important direction is to equip the models with uncertainty estimates [29]. The most serious risk of replacing SOFI with a trained model is hallucination: we ask the network to recover structure that SOFI itself does not resolve, yet in doing so the model may invent connections between microtubules that are absent from the ground truth. Per-pixel uncertainty maps could serve as a built-in sanity

check, quantifying how confident the model is about each part of the reconstruction. Such confidence information is essential before any structure visible only in the network output is treated as real.

Our motion augmentation also has a structural limitation: it applies a single global displacement to the entire field of view, translating the whole structure rather than its individual parts. Real biological motion is rarely rigid. A model trained only on globally translated sequences may therefore generalise poorly to such local, non-rigid deformation. Although, the model uses the local deformable convolution to limit this dependency on global movement, but it was not tested in practice. Future research can focus on extending the augmentation to per-region or non-rigid displacement fields, and pairing it with a DPA module that estimates spatially varying alignment.

Finally, training against higher-order SOFI targets could push the spatial resolution further. This would require simulating additional data and generating the corresponding higher-order SOFI targets, but the decoder layers can be readily adapted to estimate higher-order cumulants.

Conclusion

This thesis investigated two complementary questions: whether replacing the sequential ConvGRU fusion in SOFI-MISRGRU with a Transformer’s parallel attention fusion yields better super-resolution quality or lower latency. Secondly, whether MISRGRU can be restructured into a streaming architecture that produces one super-resolved output per incoming frame, enabling continuous real-time SR video rather than periodic batch outputs. Both questions were evaluated on a synthetic fluorescence microscopy dataset spanning twelve SNR conditions (three blinking speeds $\times$ four photon levels), using the resolution-scaled Pearson correlation (RSP) and decorrelation analysis as primary metrics to match the evaluation protocol of the foundational RESURF work [22].

TR-MISR versus MISRGRU (RQ1)

The TR-MISR architecture, adapted from the satellite SR domain to second-order SOFI upscaling ($2H-7$ output size), produces SR reconstructions that are competitive with the 20-frame MISRGRU foundational model. On fast-blinking and high-SNR conditions the two architectures perform similarly in RSP; on slow-blinking and low-SNR conditions, TR-MISR holds its quality while MISRGRU degrades more sharply and exhibits hallucinated filaments. The advantage of the attention mechanism is therefore most visible precisely where it matters most for live-cell imaging: difficult acquisitions with sparse blinking events and low photon counts.

TR-MISR does not, however, outperform MISRGRU on raw inference latency. At $N=20$ frames the lightest TR-MISR configuration (1 attention head, 1 Transformer layer; 1h1d) takes 262 ms per reconstruction compared to 170 ms for MISRGRU on the same hardware. An ablation across four Transformer sizes (1h1d through 8h6d) showed that RSP saturates immediately: quality does not improve meaningfully beyond 1h1d despite a $3\times$ increase in parameters. Latency, by contrast, grows steeply. From 88 ms to 209 ms at $N=8$ going from 1h1d to 4h3d. The encoder and decoder are the architectural bottleneck for quality, not the size of the fusion module. Future TR-MISR improvements should target deeper residual encoders or deformable convolutional decoders rather than larger Transformers.

Frame generalisation of TR-MISR

The most practically useful property of the Transformer fusion module is its length-invariance at inference time. Because the cross-attention softmax redistributes weight over all input tokens rather than accumulating magnitude, the fused representation fed to the decoder remains at a stable scale regardless of how many frames are provided.

This stability means that a TR-MISR model trained on 8 frames can be evaluated on 20 to 40 frames at test time without retraining or architectural changes, if the memory allows it. Providing 20 frames to an 8-frame-trained model matches or exceeds the RSP of the 20-frame MISRGRU baseline. A counterintuitive but robust finding is that an 8-frame-trained TR-MISR outperforms a 20-frame-trained TR-MISR when both are evaluated on 20-frame inputs. Training under the tighter 8-frame budget forces the attention module to assign discriminative weights to every available frame; training on 20 frames

risks over-saturating the cross-attention on redundant repeated blinking observations of the same fluorophores. The practical consequence is that TR-MISR can be trained on short sequences, reducing compute cost and training time, while the researcher retains the option of feeding a longer sequence at inference for improved quality.

Streaming MISRGRU and real-time deployment (RQ2)

The streaming MISRGRU variant, which emits one super-resolved frame for every input frame by carrying the ConvGRU hidden state across the entire acquisition without resetting, addresses the throughput bottleneck of the batch model. Batch MISRGRU produces one SR output per N acquired frames; the streaming model produces one SR output per frame.

At $N=20$ frames on a 512×512 field of view, the streaming model takes 11.29 ms per SR output on an NVIDIA RTX A5000 which is a $15\times$ improvement over the batch model's 170.29 ms per output at the same frame count. At 20 accumulated frames the streaming model's RSP converges to the batch model's value, and it continues to improve beyond that point as the GRU accumulates more context across the acquisition a capability the batch model lacks entirely.

Training the streaming model required truncated backpropagation through time (TBPTT) to keep memory tractable for long sequences. Supervising only the last n frames of each TBPTT chunk was critical: supervising every frame from the first step prevented the GRU from learning to defer quality improvements to later frames. An ablation across TBPTT window length, loss function, GRU width, and decoder type determined the final configuration: LLN35 supervision window, Fourier-only loss, a single 48-channel ConvGRU, and a transposed-convolution decoder.

Motion reconstruction (RQ3)

The static streaming model was further extended to handle inter-frame sample motion, which is unavoidable in live-cell imaging. Two strategies were evaluated: direct finetuning on motion-augmented data, and the addition of a Deformable Phase-space Alignment (DPA) module that corrects the GRU hidden state into the coordinate frame of the current input before the recurrent update. The final DPA model is effectively motion-invariant: at fast/high-SNR its reconstruction fidelity holds whether the sample is static or moving up to $5.0 \mu\text{m/s}$, whereas the static-trained TR-MISR and MISRGRU baselines collapse at those same speeds. DPA also outperforms the plain finetune-only variant, with visual inspection confirming less ghosting and blur from misaligned hidden states.

On static data the only degradation is in slow-low-SNR conditions, where long-range temporal averaging is most valuable and a model trained to absorb inter-frame displacement can no longer fully exploit it; here the TRMISR keep a small edge. For deployment, the unmodified streaming model is preferred when the sample is stationary or drift is sub-pixel; the DPA variant should be selected whenever the specimen exhibits visible drift relative to the field of view.

Summary and outlook

Taken together, the results confirm that deep-learning SR models can produce second-order-SOFI-quality reconstructions from as few as 8 fluorescence frames in a fraction of the time required by classical SOFI. TR-MISR's attention mechanism provides robust quality under low-SNR conditions and a practically valuable frame-count generalization property that the GRU cannot match. The streaming MISRGRU architecture removes the batch acquisition bottleneck entirely, delivering continuous SR video at 11 ms per frame or 17 ms for DPA model, comfortably within the 25 ms budget of a 40 fps acquisition, while improving the reconstruction quality with more frames input.

All quantitative results are obtained on synthetic simulated microtubule data. Transfer to real experimental microscope data remains an open step before the actual deployment. Additionally, higher-order SOFI targets (third or fourth cumulants) could be incorporated to push spatial resolution further; the decoder architecture is straightforwardly extensible to this setting.

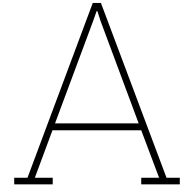
The combination of a streaming output model achieving real-time SR video and a Transformer that robustly aggregates frames under challenging imaging conditions brings deep-learning super-resolution microscopy a meaningful step closer to routine use in real-time live-cell biology.

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TR-MISR Validation Loss

Figure A.1 compares L1 Fourier loss on validation set during training. The data is taken directly from tensorboard. Despite the 20 frame model achieving lower loss compared to 8 frame model, when both models are evaluated with 20 input frames the 8 frame TR-MISR reaches better reconstruction accuracy measured with resolution scaled pearson correlation (RSP). More detail in Section 5.1.3.

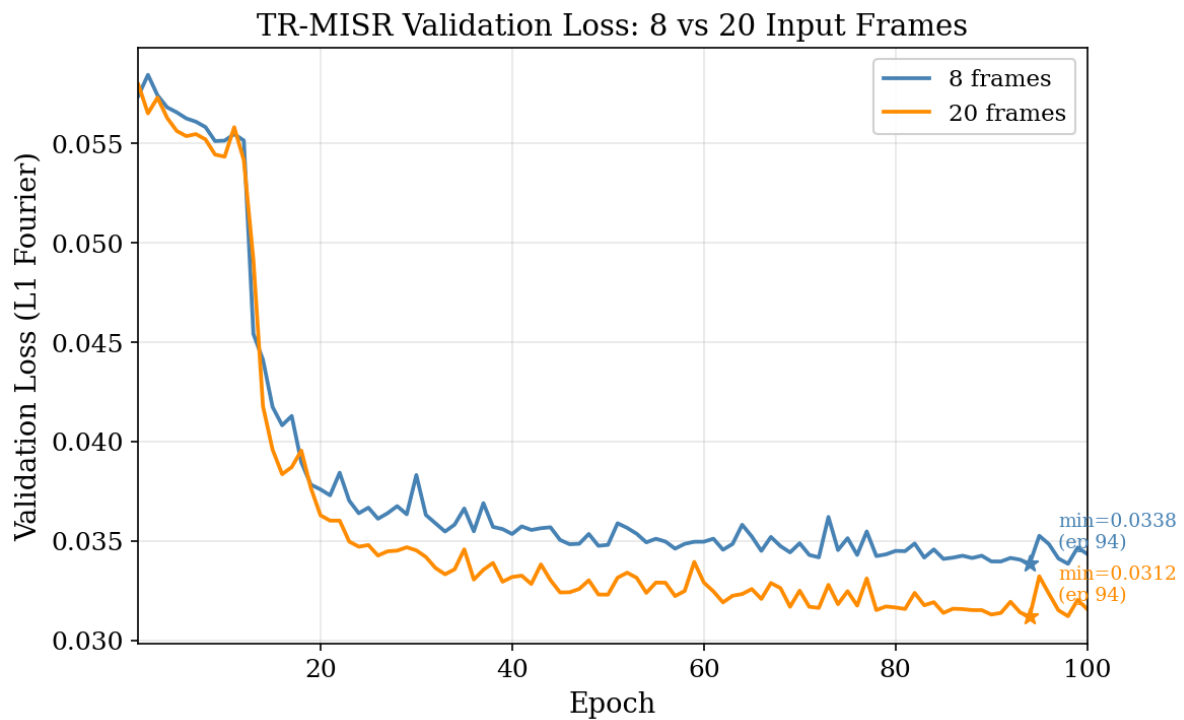


Figure A.1: Comparison of validation loss per epoch when training TR-MISR with 8 and 20 input frames.

B

TRNet Architecture and Training Ablations

This appendix reports the ablation experiments that informed the encoder, decoder and loss-function choices used in all main-chapter TR-MISR experiments (Section 5.1.1). Three design choices were evaluated independently against the chosen baseline: First, feeding the encoder a single fluorescence channel versus concatenating an averaged reference image as a second channel (Section 3.1); secondly, replacing the PixelShuffle decoder with a transposed-convolution (deconvolution) decoder; and thirdly, replacing the combined L1+Fourier loss with the Fourier-only loss. All other hyperparameters are held fixed at the values described in Section 5.1.1.

B.1. Encoder: Single-channel vs. Added Averaged Reference Image

The original TR-MISR encoder takes a two-channel input, concatenating each low-resolution frame with a reference image that acts as an implicit co-registration anchor [1]. As argued in Section 3.1, in SOFI the frame stack is co-registered by construction, so this reference is expected to be redundant. This ablation tests that claim empirically by comparing two otherwise identical 1-head, 1-depth, TR-MISR models evaluated at a 20-frame budget: the chosen single-channel model, and a two-channel variant whose second channel is the per-stack averaged reference image. We chose average over median as the background noise will get averaged out and only the diffraction limited structure remains.

The comparison is paired: both models are evaluated on the identical 480-sample test set and scores are matched by condition and sample identity. Following the procedure in Appendix E, we report the mean resolution-scaled Pearson (RSP) difference $\bar{d}$, the two-sided Wilcoxon signed-rank p -value, and the matched-pairs rank-biserial correlation.

The single-channel encoder is significantly better in every condition and on every paired sample. Pooled over all 480 samples it gains $\bar{d} = +0.0266$ RSP ($p \approx 2.3 \times 10^{-80}$, rank-biserial = -1.00). The margin grows monotonically as the imaging conditions worsen, from $+0.007$ RSP at the easiest fast/high-photon setting to $+0.068$ RSP at the hardest 12_slow_p40 condition. Concatenating an averaged reference image therefore does not help and actively hurts: the reference adds a redundant, low-fluctuation channel that dilutes the blinking statistics the fusion relies on, with the largest cost precisely where the fluctuation signal is weakest. This confirms the design choice of a single-channel encoder (Section 3.1).

B.2. Decoder: PixelShuffle vs. Transposed Convolution

The SOFI super-resolution task requires upscaling LR from $N \times N$ to HR result of size $2 * N - 7 \times 2 * N - 7$. Two standard decoder approaches were compared:

- **PixelShuffle** (sub-pixel convolution [33]): Originally used in the TR-MISR paper cite[1]. It learns to rearrange channel dimensions into spatial resolution. The non-integer scale is handled by

upsampling to an intermediate size followed by a centre crop to match the $2N - 7$.

- **Transposed convolution** (ConvTranspose2d): Is used in MISRGRU models from RESURF [22] learned upsampling via strided deconvolution. Known to produce checkerboard artefacts when the kernel size is not divisible by the stride [27].

B.3. Loss Function: L1+Fourier vs. Fourier Only

The Fourier loss [22] penalises errors in the amplitude and phase spectra of the prediction, encouraging recovery of high-frequency detail that an L1 loss would suppress via mean regression. A combined L1+Fourier loss retains the spatial stability of L1 while adding spectral supervision. The ablation tests whether the L1 term is redundant or provides complementary signal.

B.4. Results

Table B.1 reports aggregate metrics over all 480 test samples (12 SNR conditions $\times$ 40 samples each). Table B.2 gives the RSP breakdown per SNR condition.

Table B.1: Aggregate metrics for the three configurations over 480 test samples. Bold marks the best value per column.

Model	RSP
L1+Fourier, PixelShuffle (baseline)	0.8789
Fourier only, PixelShuffle	0.8757
L1+Fourier, Deconv	0.8770

Table B.2: RSP per SNR condition. Conditions are ordered by blinking speed (fast/medium/slow) and photon level (100%–40%). Bold marks the best value per row.

Condition	L1+F, PS (baseline)	F only, PS	L1+F, Deconv
01 fast p100	0.9013	0.8987	0.8995
02 fast p80	0.8996	0.8984	0.8988
03 fast p60	0.9022	0.9024	0.9021
04 fast p40	0.8950	0.8998	0.8964
05 medium p100	0.8885	0.8881	0.8872
06 medium p80	0.8900	0.8915	0.8892
07 medium p60	0.8845	0.8872	0.8839
08 medium p40	0.8843	0.8861	0.8842
09 slow p100	0.8579	0.8548	0.8545
10 slow p80	0.8536	0.8483	0.8498
11 slow p60	0.8556	0.8453	0.8510
12 slow p40	0.8345	0.8073	0.8269
Aggregate		0.8757	0.8770

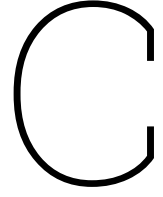
B.4.1. Decoder comparison

The Deconv decoder performs comparably to PixelShuffle at high-SNR conditions (fast blinking, high photon counts) but falls behind consistently at medium and slow blinking. At the most challenging condition (12_slow_p40), the PixelShuffle baseline achieves RSP = 0.835 versus 0.827 for Deconv, a gap of 0.008 that represents a meaningful difference in structural fidelity under low-SNR conditions.

B.4.2. Loss function comparison

The Fourier-only loss matches or slightly exceeds the baseline at fast-blinking, high-SNR conditions (conditions 03–04 and 06–08) but degrades at slow-blinking, low-photon conditions. At condition 12_slow_p40, the Fourier-only model achieves RSP = 0.807 versus 0.835 for the baseline, a gap of 0.027. The gap between baseline and Fourier-only grows with decreasing SNR.

The L1 term therefore provides complementary supervision: it penalises spatial misregistration directly, stabilising the model under challenging imaging conditions where Fourier-based supervision alone is insufficient.



Streaming MISRGRU Training Ablations

Before settling on the final streaming MISRGRU configuration reported in Section 5.3, a series of training-level ablations were conducted to identify the best combination of:

- truncated back-propagation-through-time (TBPTT) supervision window
- loss function
- ConvGRU size
- decoder architecture

Each ablation isolates one variable while keeping the others fixed. Results are reported on the same 480-sample test set and evaluated using RSP.

Following parameters were used to train baseline Streaming MISRGRU model: For TBPTT sequence of 50 frames, chunk size of 16 and loss last N 20. trained with 2000 samples with 90/10 train/val split.

Due to time limits of training on DelftBlue cluster all models were given same 24 hours limit.

Truncated BPTT: Loss-last-N

Streaming MISRGRU is trained with TBPTT over sequences of 50 frames. Computing gradients through the entire sequence is prohibitively expensive, so back-propagation is truncated to the final N frames. Loss last N parameter controls both memory cost speed of training and how broadly the hidden-state trajectory is supervised.

Figure C.1 compares RSP on the evaluation set $N = 10$ (where gradients cover only the last 20% of the sequence) and $N = 35$ (covering the last 70%).

Despite model trained with $N = 35$ achieving a highest final L1+Fourier validation loss (0.0293 vs. 0.0284 of $N = 10$), the LLN10 model scores lower on RSP. The tighter supervision window trains the GRU to produce good outputs at the tail of the sequence.

Loss function: Fourier vs. L1+Fourier

We trained two streaming models under identical conditions (with lost last N of 20, two-GRU 24-channel architecture, deconvolution decoder) but with different loss functions: the pure Fourier loss $\mathcal{L}_{\text{Fourier}}$ and the combined $\mathcal{L}_{\text{L1+Fourier}} = \mathcal{L}_{\text{L1}} + \lambda \mathcal{L}_{\text{Fourier}}$ with λ being 0.1.

Figure C.2 shows that the Fourier-only model outperforms L1+Fourier, particularly under low-SNR conditions (slow blinking, reduced photon count). The L1 term introduces a regression-to-mean bias: because SOFI images are dominated by near-zero background pixels, minimising the mean absolute error encourages the model to slightly suppress fine structure in order to reduce the background

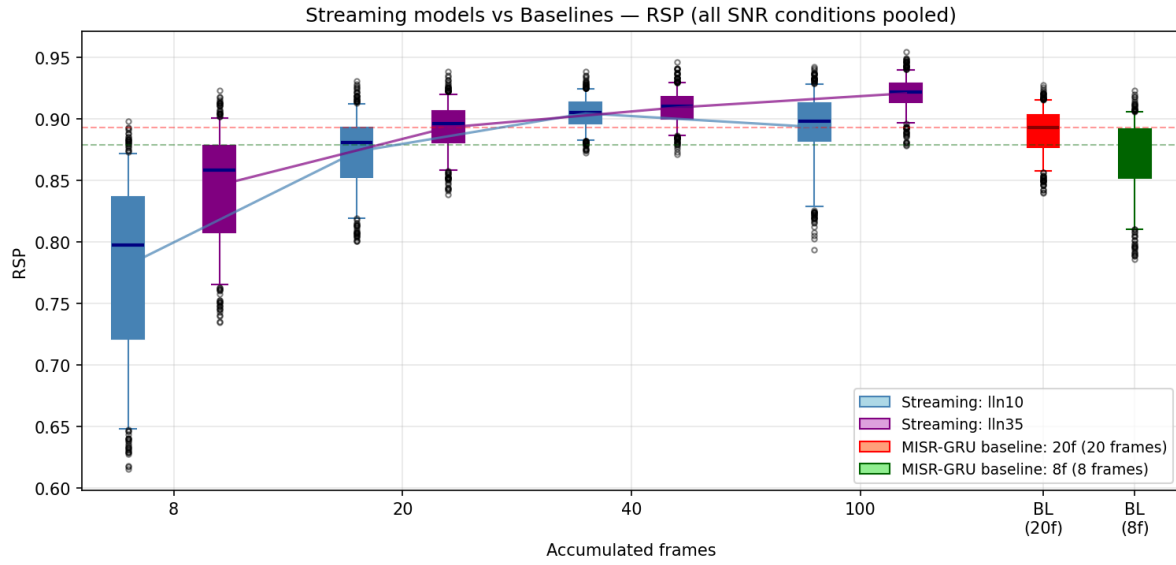


Figure C.1: RSP evaluation of models when trained with $N = 10$ and $N = 35$.

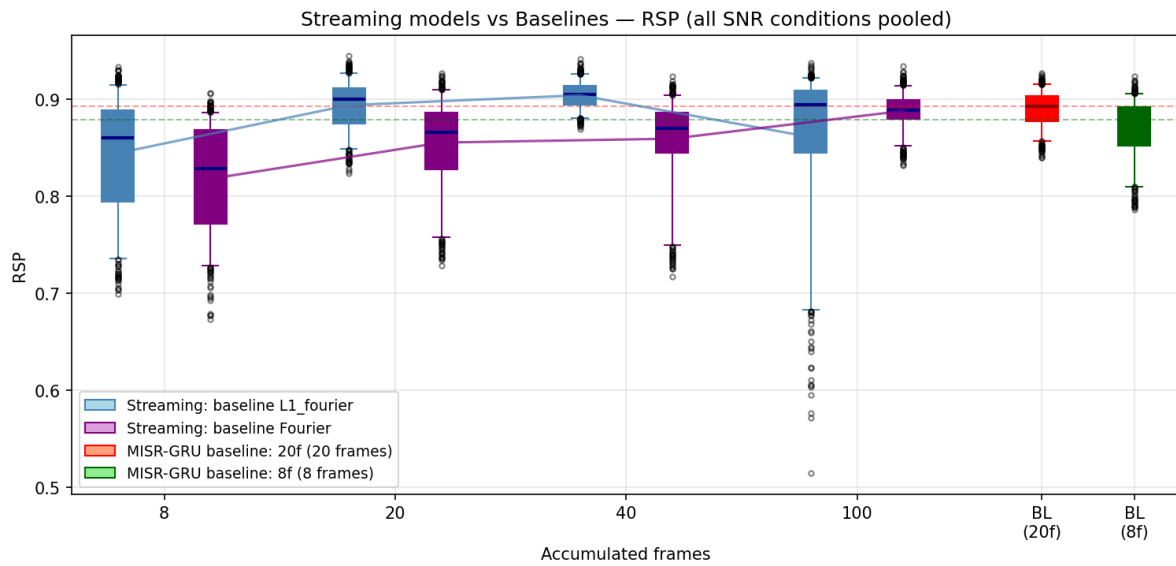


Figure C.2: RSP on the test set per SNR condition for streaming MISRGRU trained with Fourier-only loss versus combined L1+Fourier loss.

contribution to the loss. The pure Fourier objective has no such incentive; it rewards spectral fidelity at all spatial frequencies regardless of pixel intensity.

GRU architecture: two 24-channel GRUs vs. one 48-channel GRU

The original MISRGRU fusion module stacks two ConvGRU layers each with 24 feature channels. To test whether a single wider GRU with the same total channel budget can match or improve upon this design, we trained a variant with one ConvGRU of 48 channels. Both models were trained with the Fourier loss and LLN20.

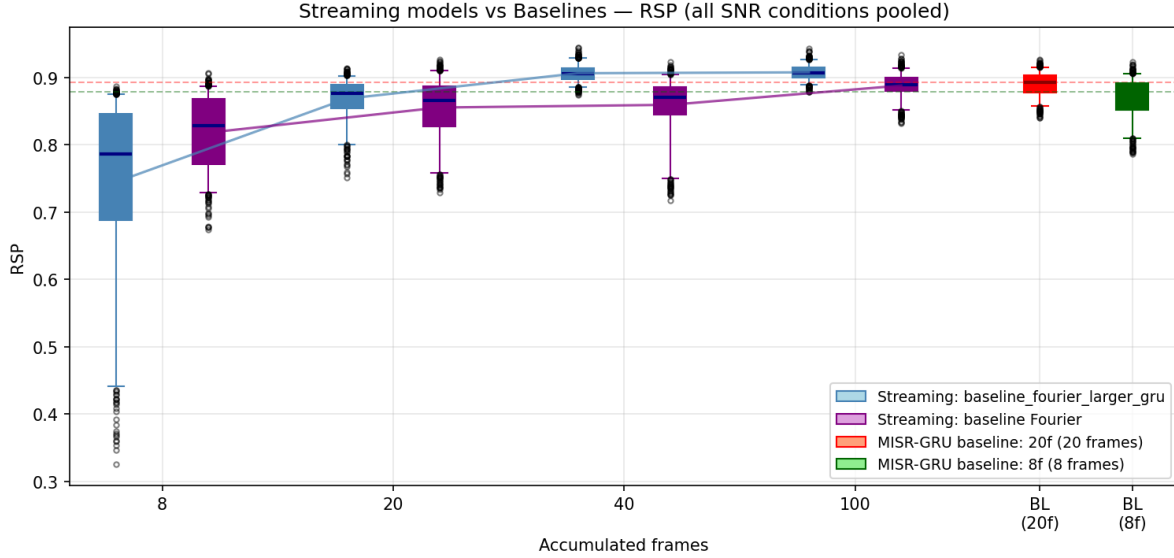


Figure C.3: RSP per SNR condition for streaming MISRGRU with two 24-channel ConvGRUs versus a single 48-channel ConvGRU, both trained with Fourier loss.

Figure C.3 shows that the single 48-channel configuration ultimately outperforms the two 24-channel baseline on RSP, despite exhibiting slower convergence during training. As visible in Figure C.4, the 1×48 model takes more epochs to reach its lowest validation loss compared to the 2×24 baseline, yet it achieves a lower final loss and a higher RSP at the end of the 24-hour training limit.

The slower convergence is expected: a single GRU with 48 channels must simultaneously learn to extract local spatial features and integrate them into a global hidden state. However, at the end of 24 hour training limit the larger hidden state of the 1×48 GRU provides richer representational capacity. Because the streaming model processes frames one at a time and must maintain all accumulated information in a single hidden state vector, a wider hidden state can retain more independent blinking-event statistics simultaneously.

Decoder: PixelShuffle vs. transposed convolution

The non-integer SOFI upscaling factor $(2H - 7) \times (2W - 7)$ requires a decoder that can produce an output of exact size without cropping artefacts at the boundary. Two designs were evaluated: a $2 \times$ PixelShuffle followed by a centre-crop to discard the seven-pixel border, and a learned transposed convolution (ConvTranspose2d) configured to output exactly $(2H - 7) \times (2W - 7)$ pixels by adjusting stride, padding, and output-padding.

Figure C.5 shows that the transposed-convolution decoder performs slightly better than PixelShuffle on all pooled SNR conditions. We believe the issue comes from the cropping to match $2n-7$ upscaling, while transposed convolution maps directly to the exact $(2H - 7) \times (2W - 7)$ output grid and avoids this issue.

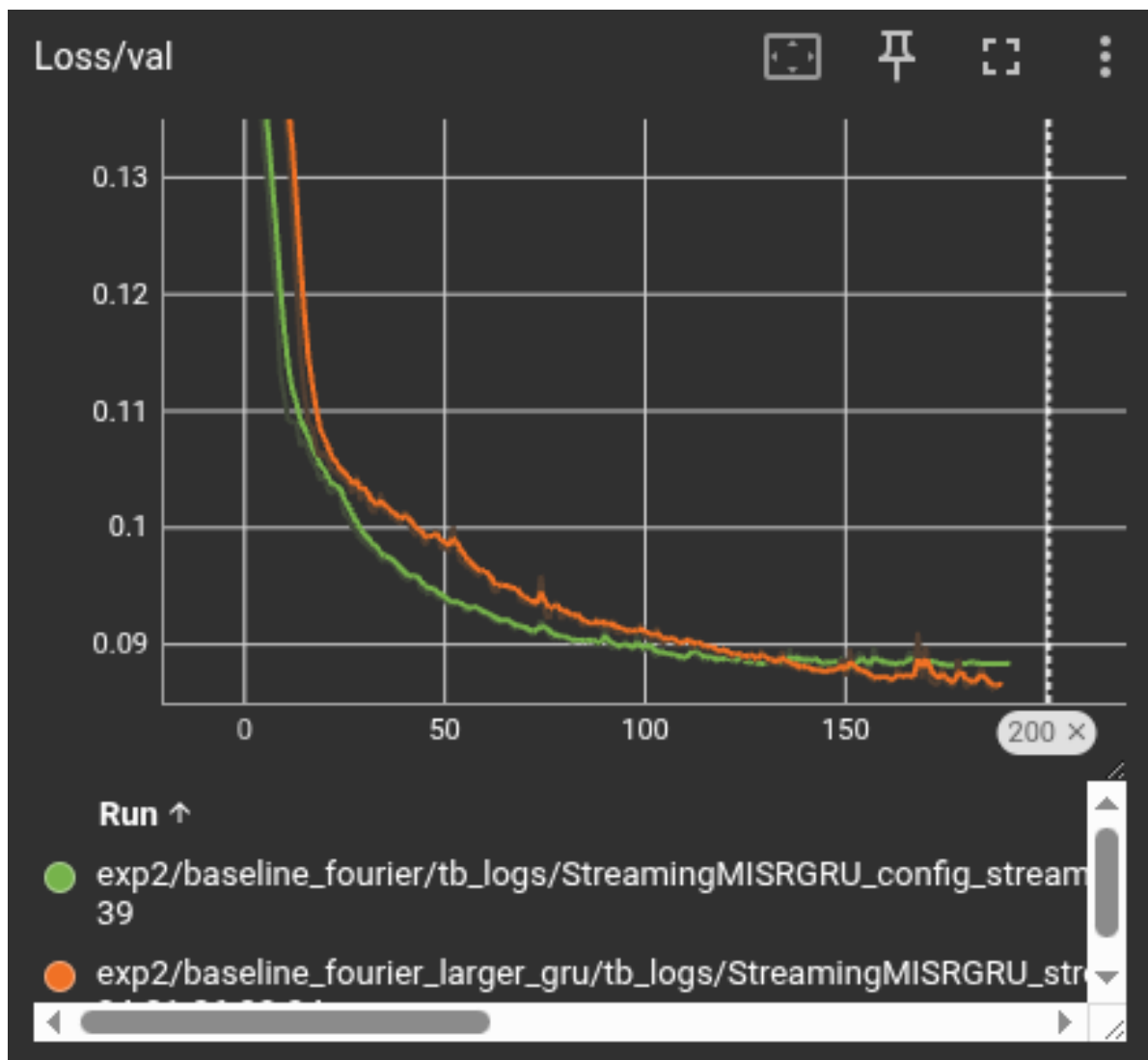


Figure C.4: Validation loss during training of 1x48 ConvGRU vs 2x24 ConvGRUs

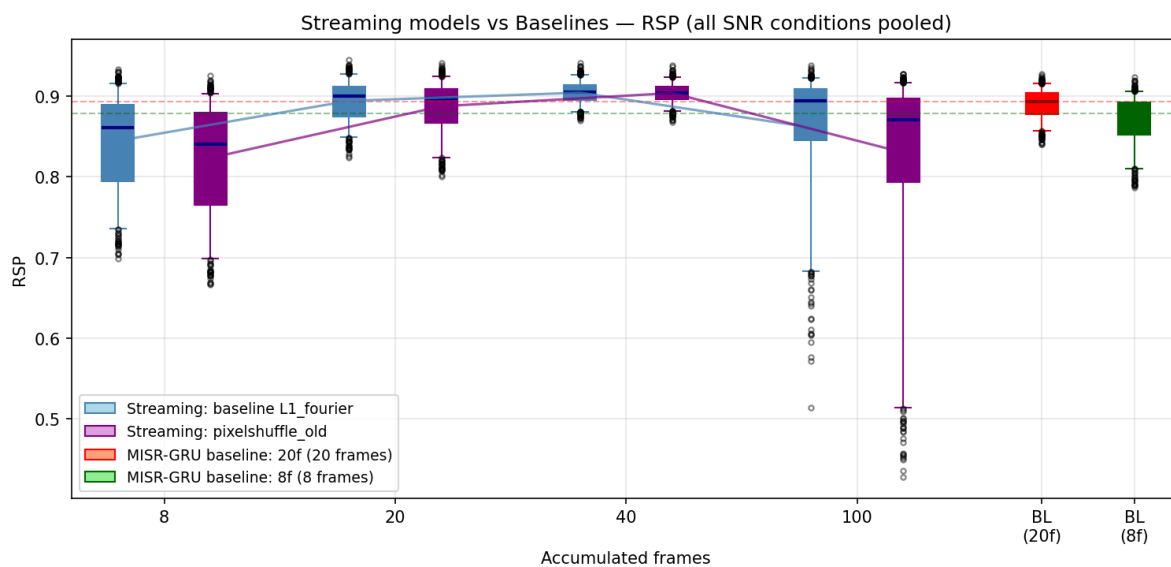


Figure C.5: RSP per SNR condition for streaming MISRGRU with a PixelShuffle decoder versus a transposed-convolution decoder.

D

Validation Loss Across Transformer Sizes

All four runs converge to essentially the same floor, with best values of 0.0329–0.0332. Larger models descend faster per epoch but reach only slightly lower asymptote, while the lightweight 1h1d configuration matches them given enough epochs.

The training budgets were not equal. All runs were limited to 24 h on DelftBlue, and only the 1h1d configuration completed the full 400 epochs (in 14 h); the 2h2d, 4h3d and 8h6d runs timed out at progressively fewer epochs (360, 266, 140 respectively) because each epoch is more expensive for the larger models. That the larger variants matched 1h1d *despite* receiving less training reinforces the conclusion that the fusion module saturates at minimal capacity, and that the preferred 1h1d configuration sacrifices no reconstruction accuracy.

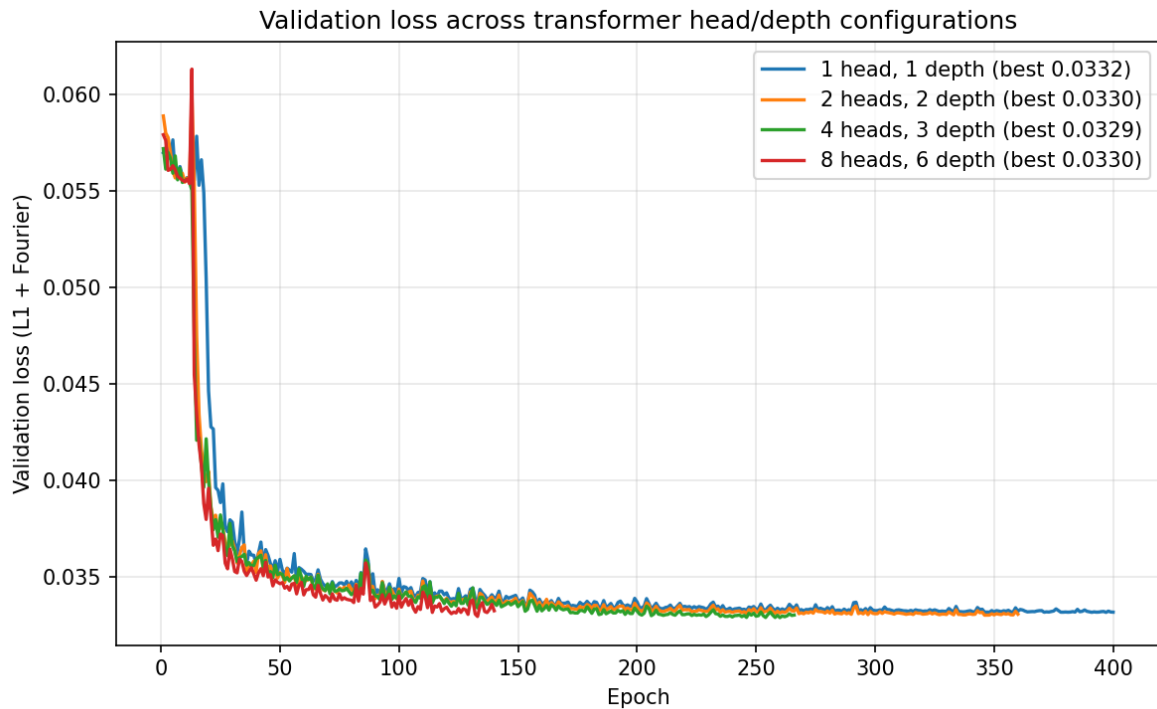


Figure D.1: Validation loss (L1 + Fourier) per epoch during training for the four Transformer head/depth configurations. All converge to nearly identical loss. Runs were limited to 24 h, so the larger variants completed fewer epochs than 1h1d.

Statistical Significance of Model Comparisons

This appendix reports the statistical tests behind the model comparisons in Section 5.1.2. Three fusers are compared on the static test set: the recurrent baseline **MISRGRU** [22], the transformer fuser **TR-MISR** (8-frame), and the **DPA** model (streaming MISRGRU with the Deformable Phase-space Alignment module).

Procedure

All comparisons are paired. Every model is evaluated on the identical 480-sample test set (12 blinking-speed $\times$ photon-level conditions, 40 samples each), and scores are matched by condition and sample identity. For a pair of models A and B we form the per-sample differences of the resolution-scaled Pearson correlation score, $d_i = \text{RSP}_i^A - \text{RSP}_i^B$, and test the null hypothesis that their median is zero.

Because RSP is bounded in $[-1, 1]$ and not normally distributed, the primary test is the two-sided **Wilcoxon signed-rank test**. When several pairs are compared within one family of conditions, the p -values are adjusted for multiple comparisons with the **Holm-Bonferroni** method, and significance is declared at an adjusted level of $\alpha = 0.05$.

We also show the **mean RSP difference** $\bar{d}$, and the **matched-pairs rank-biserial correlation** $r \in [-1, +1]$ measures how consistently one model wins ($r = +1$ means A beats B on every sample, $r = 0$ means wins and losses balance). Because the sample sizes are large ($n = 40$ per condition, $n = 480$ pooled), the p -values are extremely small and several saturate the exact-test floor; interpretation therefore relies on the effect sizes rather than on the p -values themselves.

Three-model comparison at a 20-frame budget

At a matched budget of 20 input frames we compare all three models pooled over the full test set ($n = 480$).

Table E.1: Paired comparison of the three fusers at a 20-frame budget, pooled over all 480 test samples. $\bar{d}$ is the mean RSP difference ($A - B$); p is Holm-adjusted Wilcoxon. Higher RSP is better.

Comparison (A vs. B)	mean A	mean B	$\bar{d}$	better	rank-biserial
MISRGRU vs. TR-MISR	0.8895	0.9025	-0.0129	TR-MISR	-0.81
MISRGRU vs. DPA	0.8895	0.9024	-0.0128	DPA	-0.86
TR-MISR vs. DPA	0.9025	0.9024	+0.0001	-	-0.06

Both TR-MISR and DPA significantly outperform the MISRGRU baseline ($p \approx 3.1 \times 10^{-52}$ and 8.2×10^{-60} respectively), with large and highly consistent effects (rank-biserial ≈ -0.8 ; MISRGRU loses on the large

majority of samples). The final streaming MISRGRU (Section 5.3) was evaluated with the same paired procedure and likewise beats the batch MISRGRU baseline at a matched 20-frame budget ($\bar{d} = 0.010$ RSP in its favour, rank-biserial -0.99 , $p \approx 6 \times 10^{-78}$). It wins on almost every paired sample. In contrast, TR-MISR and DPA are statistically indistinguishable in aggregate: the mean difference is $+0.0001$ RSP and the Wilcoxon test is not significant ($p = 0.24$).

This aggregate tie is misleading. When we compare the models per imaging setting it reveals a clean difference: DPA is significantly better in the fast-blinking / high-SNR conditions ($\bar{d}$ from -0.015 to -0.024 in favour of DPA), whereas TR-MISR is significantly better in the slow-blinking / low-SNR conditions, its advantage growing as the signal weakens (up to $+0.040$ RSP at the hardest 12_slow_p40 condition).

DPA vs. TR-MISR at a 30-frame budget

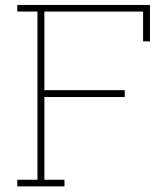
Increasing the number of input frames to 30 frames for both models confirms the same pattern. Pooled over all conditions ($n = 480$), DPA holds a small but significant edge ($\bar{d} = -0.0034$ RSP, $p = 2.9 \times 10^{-6}$, rank-biserial -0.25), a tie that hides a tradeoff.

Table E.2: DPA vs. TR-MISR at a 30-frame budget, grouped by blinking speed. $\bar{d}$ is the mean RSP difference (TR-MISR – DPA); ★ marks Holm-adjusted Wilcoxon $p < 0.05$.

Condition group	$\bar{d}$ (TR-MISR – DPA)	better	rank-biserial	sig
fast (p100–p40)	-0.019 to -0.023	DPA	-1.00	★
medium (p100–p60)	-0.006 to -0.012	DPA	-0.99 / -1.00	★
medium p40	$+0.001$	— (n.s.)	$+0.30$	
slow (p100–p40)	$+0.008$ to $+0.032$	TR-MISR	$+1.00$	★

The sign of the difference flips monotonically with blinking speed: under fast blinking DPA wins every sample (rank-biserial $= -1.0$), the medium conditions form the crossover region, and under slow blinking TR-MISR wins every sample, with its margin increasing as the photon count drops (up to $+0.032$ RSP at 12_slow_p40). The photon level does not change which model wins but modulates the size of the gap.

In summary, at equal frame budgets both fusers clearly beat the recurrent MISRGRU baseline, but neither dominates the other: DPA is the stronger model under fast-blinking / high-SNR conditions, TR-MISR under slow-blinking / low-SNR conditions, the two crossing over in the medium regime. Because these opposing effects cancel when conditions are pooled, the comparison is only meaningful when stratified.



Disclosure of AI Usage

Following the TU Delft guidelines on the responsible use of Artificial Intelligence, this appendix transparently documents how AI tools were employed throughout the research and writing of this thesis:

- **Literature Discovery:** Claude and ChatGPT were used for the initial discovery of relevant academic literature and sources. All sources identified by these AI tools were subsequently verified before inclusion in this thesis.
- **Code Development and Debugging:** Claude Code was used as a programming assistant to help with writing scripts for running evaluations and creating figures, debugging, and understanding codebase during the development of this thesis.
- **Text Editing and Rephrasing:** Claude and ChatGPT were used for proofreading, grammatical corrections, and rephrasing certain sections of the text to improve overall clarity, readability, and flow.
- **LaTeX formatting:** Claude Code assisted with structuring and formatting the document, including the layout of tables and figures.
- **Conceptual Learning:** Claude and ChatGPT were used to learn and clarify domain concepts encountered during the research. All explanations were cross-checked against the primary literature.
- **Image Creation:** Gemini assisted with the creation of two figures describing the model architecture to follow the same style as in the original papers.